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Research Paper

Unlocking mid-to-long-term flexibility: why seasonal pumped storage outperforms conventional pumped storage in wind-solar dominated grids

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ABSTRACT

Global efforts toward Dual Carbon Goals have spurred rapid growth in wind and solar installations worldwide. However, the inherent randomness and intermittency of wind and solar pose critical challenges to long-term flexibility requirements in power systems. While pumped storage remains a crucial regulating power source, the differential effectiveness between seasonal pumped storage (SPS) and conventional pumped storage (CPS) in mitigating medium-to-long-term renewable fluctuations has not been systematically investigated. Therefore, this study provides a comprehensive evaluation of the performance of SPS and CPS in addressing stochastic fluctuations of wind and solar. First, the output characteristics of wind and solar resources were analyzed, and multiple wind and solar scenarios were developed to simulate the regulatory differences between SPS and CPS under varying wind and solar scenarios. Subsequently, this study proposes an integrated evaluation framework combining entropy-weighted and coefficient of variation methods to objectively assess the technical, economic, and stability indicators, using Qinghai Province as a case study to analyze the advantages of both SPS and CPS. The results show that SPS outperforms CPS across many technical indicators. Notably, the carbon emission reduction effect of SPS becomes more pronounced in scenarios with higher wind power capacity. Economically, it has to be admitted that the investment cost of SPS is 1.33 times that of CPS. Although the operating cost is one fifth of CPS, the levelized cost of electricity of SPS is still higher than CPS. Finally, the comprehensive evaluation demonstrates the superiority of SPS, achieving an overall benefit score of 98.03, in stark contrast to 68.6 for CPS. These findings offer critical insights for energy planners and policymakers in optimizing storage solutions to enhance grid flexibility and accelerate decarbonization.

1. Introduction

1.1. Background

Global carbon neutrality goals are driving an unprecedented acceleration in renewable energy deployment, fundamentally reshaping the world's energy landscape [1,2]. In 2023 alone, global renewable capacity additions surged by nearly 50 % to almost 510 GW, with wind and solar power accounting for the vast majority [3,4]. China has been the primary engine for this growth, the country's combined wind and solar capacity reached 1,250 GW by 2024, meeting its Climate Summit commitments six years early [5,6]. However, increasing renewable penetration creates operational challenges due to wind and solar variability, threatening power system stability. The primary challenge has shifted from short-term power balance to mid-to-long-term flexibility requirements, demanding new operation and dispatch strategies. Seasonal pumped storage (SPS) has emerged as a critical solution, providing unique mid-to-long-term energy regulation capabilities essential for enhanced system flexibility.

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1.2. Problem statement

Power systems in regions with abundant wind and solar resources exhibit pronounced seasonal variability due to their dependence on meteorological conditions [7]. Seasonal generation becomes highly uncertain. Similarly, conventional hydropower, which relies on seasonal runoff, faces significant fluctuations in generation capacity and regulatory potential between wet and dry seasons. To maintain a supply–demand balance, reservoirs must store water during wet seasons and allocate it strategically during dry periods. However, as the share of installed renewable capacity grows, the limited flexibility of conventional reservoir regulation may lead to a critical shortage in hydropower's mid-to-long-term compensatory capacity.

Conventional pumped storage (CPS), while useful for short-term peak shaving and frequency regulation, is fundamentally constrained by its limited storage scale. This renders it incapable of providing the sufficient long-duration discharge capacity needed to bridge the widening gap in mid-to-long-term flexibility created by high VRE penetration. In contrast, easonal Pumped Storage (SPS) has emerged as a critical solution specifically designed to address this long-term flexibility deficit. The feasibility of this concept is firmly grounded in the operational reality of Europe's key hydropower nations. For instance, Norway's extensive reservoir network demonstrates the technical maturity of large-scale hydro storage [8]. While the Swiss system's seasonal balancing function underscores the strategic value and proven viability of such long-duration assets [9].Despite its advantages, systematic research comparing the effectiveness of SPS and CPS in mitigating the stochastic fluctuations of wind and solar power remains limited. Given these challenges, as shown in Fig. 1, this study addresses the following critical scientific questions:

(i) What methodological framework can be developed to optimize power systems while accurately reflecting the stochastic and seasonal nature of wind and solar generation?
(ii) How can the comprehensive benefits of SPS be scientifically quantified under stochastic fluctuations of wind and solar resources, and what fundamental distinctions differentiate SPS from CPS?

1.3. Literature review

SPS, an emerging energy storage technology with mid-to-long-term regulation capabilities, has garnered significant attention from both academic and industrial sectors in recent years. This paper systematically reviews existing research on SPS, focusing on three critical dimensions: potential assessment, system benefits, and evaluation methodologies.

Regarding the potential assessment of SPS, Hunt et al. (2020) conducted the first global site-selection evaluation based on economic viability. It revealed that regions with mountainous terrain hold the greatest potential for SPS deployment [10,11]. Building on this work, Hunt et al. (2021) performed an in-depth analysis of SPS projects in the Indus River Basin. The subsequent findings demonstrated SPS's distinct advantages in both operational lifespan and installed capacity [12,13]. Livia et al. (2020) evaluated the feasibility of SPS in Norway and its contributions to power systems from a technical perspective [12]. Furthermore, Hunt et al. (2017) and Mongird et al. (2019) conducted quantitative analyses of the investment costs of SPS and CPS. These studies provided foundational data for subsequent economic evaluations of SPS [14,15]. Li et al. conducted a site selection study for pumped storage in Northwest China, proposing a four-stage framework that identifies and evaluates sites with diverse topographical configurations. This approach addresses siting challenges under multiple constraints [16,17].

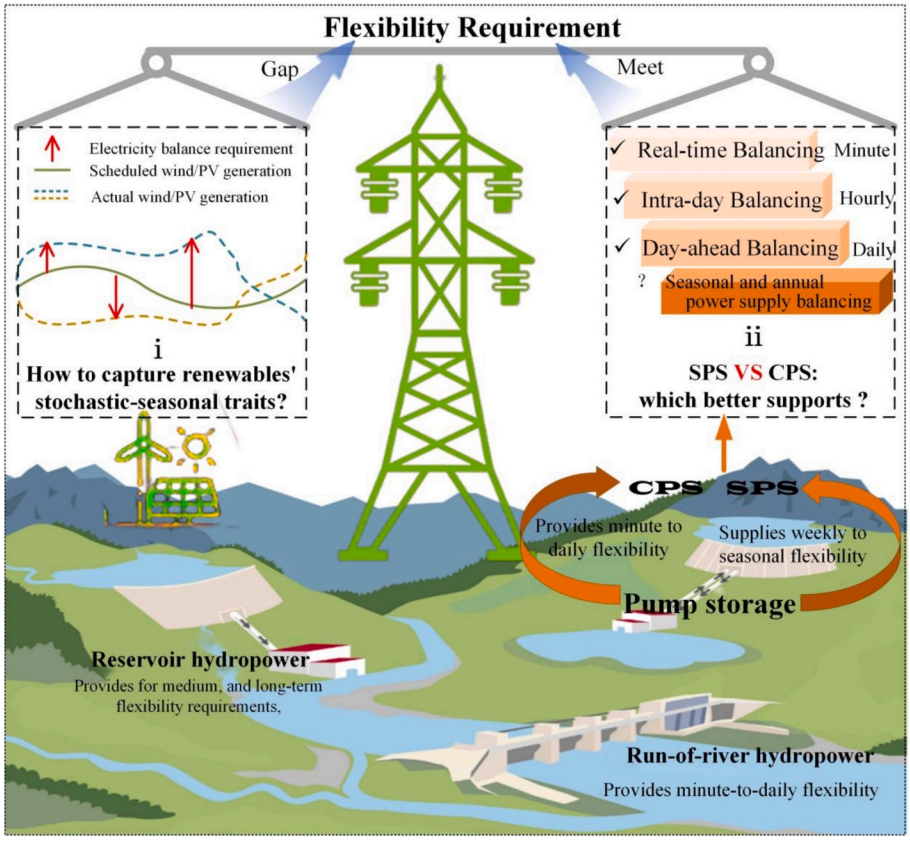


Fig. 1. Hydropower flexibility: key features and research challenges.

For system benefits, Hunt et al. (2018) evaluated SPS contributions to Brazil's power system dispatch cost optimization, demonstrating exceptional integrated economic performance [18]. Thomas et al. (2020) developed a methodology to characterize pumped storage energy flexibility, finding approximately 16 % energy savings and 57 % reduction in pumping cycles without differentiating SPS and conventional pumped storage for mid-to-long-term regulation [19,20]. Arthur et al. (2022) and Arne et al. (2017) analyzed energy storage contributions to power system flexibility amid wind and solar fluctuations, validating pumped storage effectiveness in capacity stabilization and load smoothing, but focused primarily on short-term rather than mid-to-long-term regulation [21]. Morimoto (2013) analyzed Brazil's Belo Monte Dam case, demonstrating that neglecting comprehensive economic, ecological, and social evaluations risks severe environmental consequences, providing critical guidance for sustainable SPS development [22]. Recent studies have further expanded the application scope and deepened the understanding of SPS's systemic value. For instance, some scholars have evaluated the potential of SPS in decarbonizing the power sectors of non-traditional hydropower regions, such as Saudi Arabia, highlighting its crucial role in integrating massive solar resources [23]. Others have focused on the operational optimization of hybrid systems, with Yue et al. developing a dispatch optimization model for pumped storage-wind-photovoltaic systems that explicitly considers seasonal factors to maximize overall efficiency [24]. Furthermore, the synergistic operation of SPS with existing hydropower assets has been a key research area. Studies have shown that coordinating SPS with cascade hydropower stations can significantly enhance the system's capacity to accommodate variable renewable energy [25]. The strategic value of SPS in enhancing grid resilience has also been demonstrated, particularly in its role in responding to climate change impacts, such as mitigating the effects of droughts on hydropower-dominant systems like Brazil's [26].

In comprehensive benefit evaluation methodologies, He et al. developed a multi-dimensional framework assessing cascade hydropower-renewable systems across economic, technical, and energy efficiency dimensions [27]. Zhang et al. proposed stochastic entropy-TOPSIS integration to quantify pumped storage flexibility regulation potential [28], while Zhao et al. employed weighted sum methods to evaluate operational distinctions among five pumped storage technologies [29]. Li et al. developed frameworks for assessing off-river pumped storage with floating solar systems [30] and hybrid evaluation models integrating fuzzy Delphi and entropy-weighted methods across functional, financial, and environmental dimensions [31]. Zhang et al. created comprehensive evaluation frameworks for wind-solar-pumped storage hybrid systems under varying capacity scenarios, integrating techno-economic, financial, and socio-environmental indicators [32]. You et al. proposed hybrid multi-criteria decision-making methods combining Fuzzy Best-Worst and Fuzzy TOPSIS approaches to evaluate pumped storage performance across economic, social, and environmental dimensions [33]. At the algorithmic level, Ismkhan et al. developed iterative improved k-means algorithms addressing centroid sensitivity through cluster removal and splitting [34], while Aggarwal et al. integrated nature-inspired optimization algorithms with k-means++ clustering to overcome initial centroid sensitivity and local optima limitations [35].

1.4. Research gaps and contribution

The literature review reveals substantial SPS research progress across three dimensions: potential assessment has established global site-selection frameworks identifying mountainous regions as optimal locations; system benefits research has demonstrated dispatch cost optimization and energy savings up to 16 %, though focused primarily on short-term applications; evaluation methodologies have developed multi-criteria frameworks. However, existing research has the following three gaps:

(i) Current research predominantly focuses on short-term flexibility regulation, yet a systematic investigation into the unique advantages and roles of SPS in addressing mid-to-long-term stochastic fluctuations of wind and solar resources is lacking. Furthermore, existing studies have not established systematic comparisons between SPS and CPS in mid-to-long-term regulation capacities.

(ii) While existing studies have touched upon economic or technical indicators, they often employ single-dimensional assessments. There is a notable absence of holistic, multi-dimensional frameworks that integrate critical system stability factors alongside technical and economic aspects. This oversight prevents a comprehensive understanding of SPS's value proposition, particularly in its ability to support grid stability under increasing renewable penetration.

Addressing these research gaps, this study makes the following key contributions:

(i) **Systematic quantification of mid-to-long-term flexibility from SPS and CPS**: this study proposes a comprehensive approach that systematically quantifies the distinct contributions of SPS and CPS to mid-to-long-term flexibility. By integrating a wind-solar installed capacity scenario analysis framework (leveraging the k-means++ clustering algorithm to capture distinct temporal output characteristics) with a robust power system optimization model, we enable precise and robust analysis of their fundamental differences in regulating stochastic wind and solar resources.

(ii) **Establishment of a multi-dimensional evaluation system for holistic assessment**: To overcome the limitations of single-dimensional evaluations, we construct a novel and integrated indicator system encompassing technical, economic, and unit stability perspectives. This comprehensive evaluation system, applied within our analytical framework, facilitates a holistic assessment of the intricate trade-offs and synergistic benefits of SPS and CPS.

The paper is structured as follows: Section 2 introduces the research methodology, including wind and solar output characteristic extraction techniques, power system unit commitment model with SPS, and the construction of a comprehensive evaluation framework; Section 3 presents a case study of Qinghai Province with systematically classified wind and solar resource scenario combinations; Section 4 comparatively analyzes the performance differences between SPS and CPS across multiple dimensions and provides a comprehensive evaluation; Section 5 summarizes the research conclusions and proposes policy recommendations.

2. Methodology

The technical roadmap of this study is illustrated in Fig. 2. Section 2.1 employs the k-means++ clustering algorithm to characterize uncertainties in renewable energy generation, identifies typical wind and solar output scenarios, and analyzes seasonal correlation characteristics among wind and solar. Section 2.2 develops a regional power system optimization model incorporating SPS. Section 2.3 constructs a multi-dimensional benefit evaluation framework across technical, economic, and stability dimensions, applying the entropy-weighted and coefficient of Variation method to quantitatively assess the multi-dimensional value of SPS in power systems with high renewable energy penetration.

2.1. Uncertainty characterization of renewable

To scientifically evaluate the output characteristics of wind and solar resources, this section employs the k-means++ clustering algorithm to

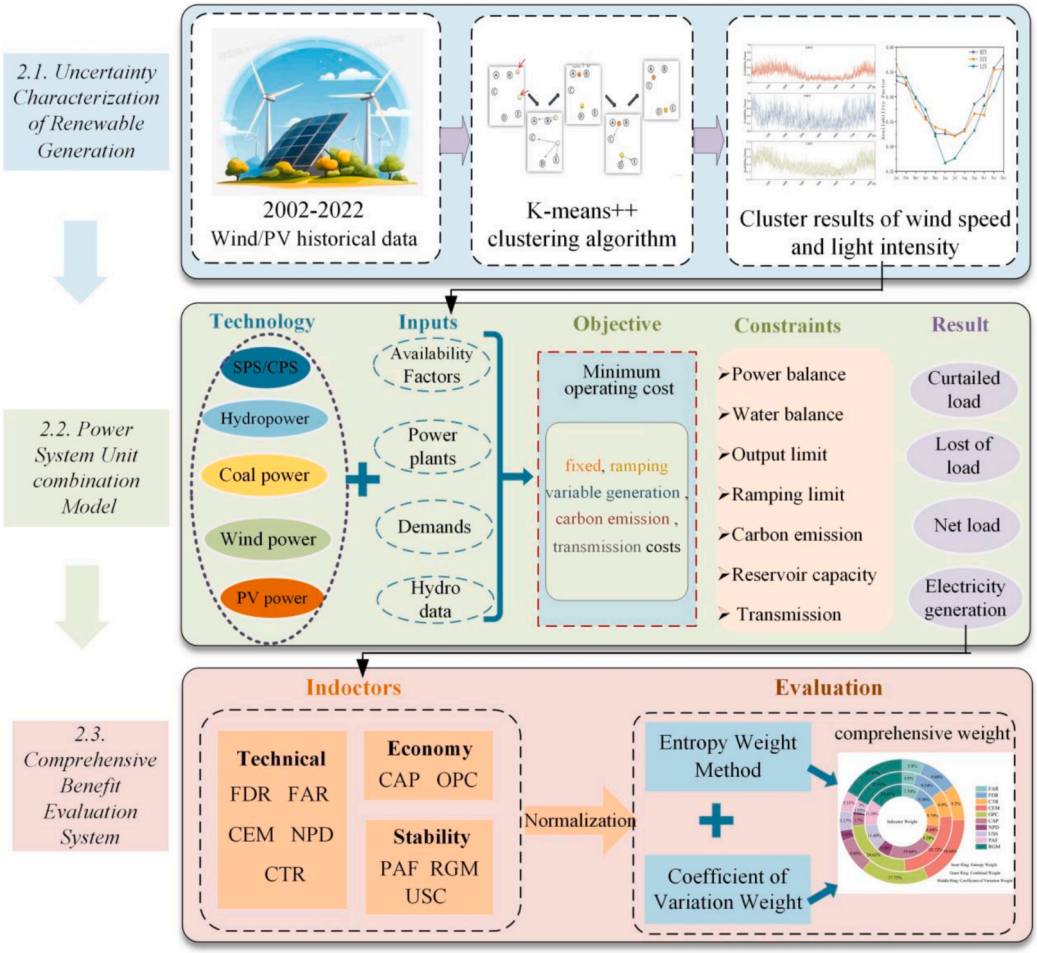


Fig. 2. Flowchart of the methodology section.

extract features from historical data. Specifically, the k-means framework is highly effective for partitioning large-scale time-series data and produces highly interpretable centroids that directly represent "typical scenarios." Furthermore, we employ the k-means++ variant to ensure robust and stable clustering results by optimizing the initial centroid selection, which overcomes a key limitation of the standard k-means algorithm [36,37].

2.1.1. *Wind/Solar output scenario clustering via k-means++*

Building on the classical k-means algorithm, the enhanced the k-means++ method [38] introduces a probabilistic initialization strategy. The first cluster center is randomly selected, while subsequent centers are chosen based on squared distance probabilities from existing centers to ensure uniform initial distribution. After initializing N centers, deterministic iterations of standard k-means proceed until convergence. This approach significantly enhances clustering stability for wind-solar scenarios Fig. 3 the specific steps are as follows:

Step 1: Initial centroid selection

The process is initiated by randomly selecting the first cluster centroid c_1 from the wind-solar time-series dataset X. The squared Euclidean distance $D(x_i)$ between each sample x_i and its nearest existing centroid is calculated. Subsequent centroids are iteratively selected based on the probability distribution formula (1), which prioritizes samples farther from existing centroids. This procedure repeats until the full set of centroids c_1 c_2 c_3 ….…c_j is obtained.

$$P(x_i) = \frac{\boldsymbol{D}(x_i)^2}{\sum_{j=1}^{m} D(x_i)^2} \tag{1}$$

where x_i is the feature vector of the *i-th* sample ; $D(x_i)$ denotes the squared Euclidean distance from the sample to its nearest cluster centroid; m indicates the total number of samples.

Step 2: Data partitioning.

For each sample, calculate its distance to all cluster centroids c_j and assign it to the nearest cluster.

$$s_j = \arg_j \min \left\| x_i - c_j \right\|^2 \tag{2}$$

where s_j denotes the cluster label of x_i; c_j represents the centroid coordinates of the *j-th* cluster.

Step 3: Centroid update.

Recalculate the centroid coordinates for each cluster and update c_j^{new} as the new cluster centroids.

$$c_j^{new} = \frac{1}{|S_j|} \sum_{x \in s_j} x \tag{3}$$

Step 4: Convergence criterion

Repeat Steps 2–3 until convergence:

$$\max_j \left\| c_j^{(t+1)} - c_j^{(t)} \right\| < \varepsilon \tag{4}$$

where c denotes the *j-th* centroid in the t-th iteration; ε represents the convergence threshold ($\varepsilon = 10^{-5}$ a predefined value).

2.1.2. *Seasonal correlation analysis of wind-solar*

This section applies the k-means++ clustering algorithm to historical wind and solar output time-series data from Qinghai Province, with

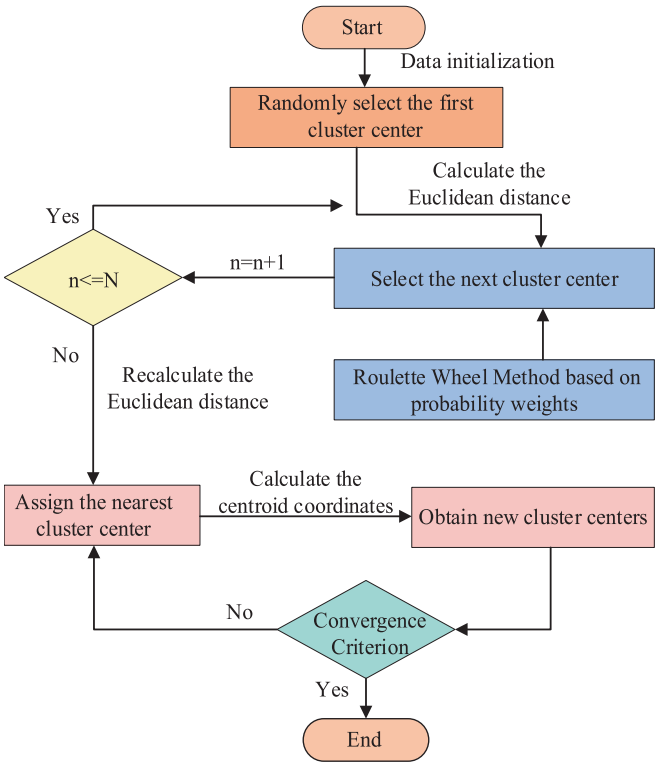


Fig. 3. Flow chart of the running steps of the k-means++ algorithm.

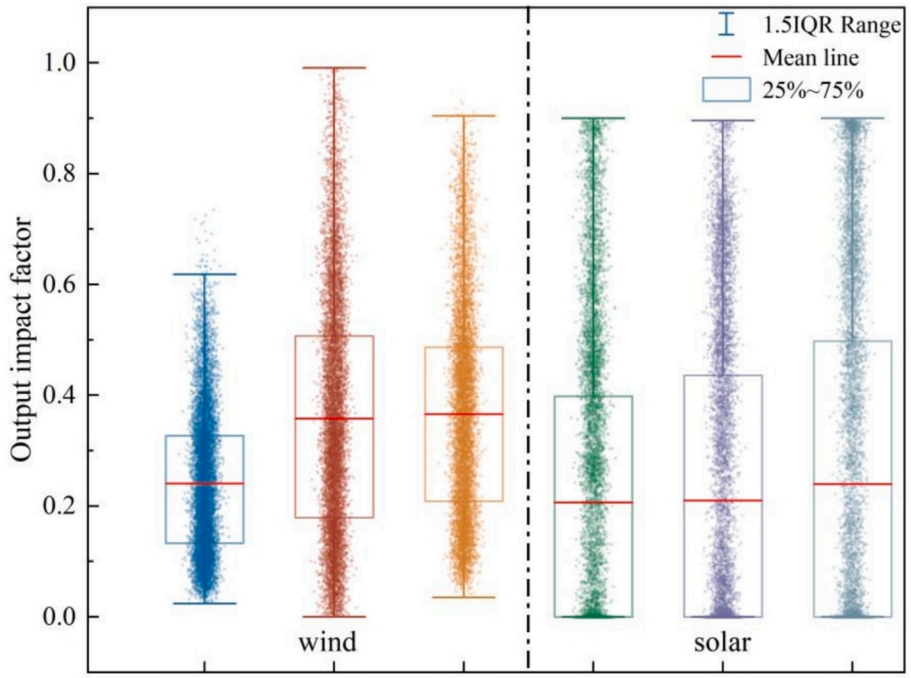
Population Mean (μ), Variance (σ^2), Standard Deviation (SD), and coefficient of variation as feature indicators. The clustering results are used to construct representative wind-solar output scenarios, capturing their typical generation characteristics.

Through iterative simulations using the k-means++ algorithm [34,38], typical scenarios of wind-solar output time-series were derived. Based on the mean and coefficient of variation of output influence factors, wind is categorized into High-Wind Year (HWY), Normal-Wind Year (NWY), and Low-Wind Year (LWY), while solar are classified as High-Irradiance Year (HIY), Normal-Irradiance Year (NIY) and Low-Irradiance Year (LIY). Specific parameters are shown in Fig. 4.

High-Wind Year is characterized by the highest mean annual output of 0.366 and coefficient of variation of 0.51, indicating sustained high output levels with minimal fluctuations. Normal-Wind Year exhibits a moderate mean output of 0.357 but the highest coefficient of variation of 0.607, reflecting significant volatility with alternating high and low output periods. Low-Wind Year shows the lowest mean output of 0.24 and a coefficient of variation of 0.533, suggesting stable low-level output with smaller fluctuations compared to Normal-Wind Year.

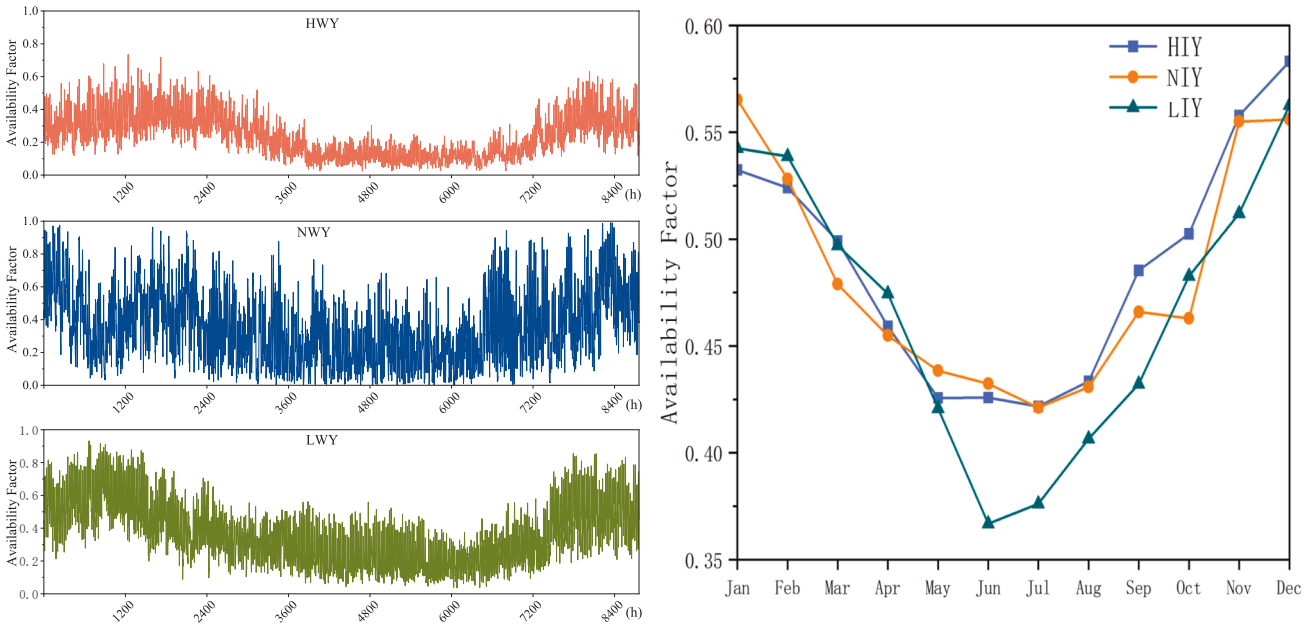
High-Irradiance Year exhibits the highest mean solar output of 0.247 with a coefficient of variation of 1.265, lower than that of Normal-Irradiance Year. This indicates sustained high solar output levels with minimal annual fluctuations. Normal-Irradiance Year shows the highest coefficient of variation of 1.274, with a mean output intermediate between High-Irradiance Year and Low-Irradiance Year, reflecting significant variability in solar generation. Low-Irradiance Year demonstrates the lowest mean output of 0.237 and a coefficient of variation of 1.269, closely resembling Normal-Irradiance Year's variability but maintaining consistently low output levels throughout the year.

The clustering results are shown in Fig. 5. To further characterize the three typical solar irradiance scenarios, the monthly effective mean irradiance intensity—calculated as the total monthly irradiance divided



	LWY	NWY	HWY	LIY	NIY	HIY
μ	0.240	0.357	0.366	0.237	0.244	0.247
σ^2	0.016	0.047	0.035	0.091	0.096	0.097
SD	0.128	0.217	0.187	0.301	0.310	0.312
CV	0.533	0.607	0.510	1.269	1.274	1.265

Fig. 4. Analysis of typical wind power output impact factor curve results.



(a) Annual wind power output factor

(b) Annual Solar power output factor

Fig. 5. Typical scenario results of wind power and solar power output of throughout the year.

by the number of daylight hours with effective irradiance–is analyzed in Fig. 5(b) This indicator accounts for the diurnal pattern of solar radiation (high intensity during daylight and zero at night).

As shown in Fig. 5(a), wind power output in Qinghai Province exhibits distinct seasonal characteristics: higher output levels prevail from January to April and October to December, while lower levels persist from May to September. Notably, high-wind years and low-wind years demonstrate pronounced seasonal variations in output patterns. As shown in Fig. 5(b), the monthly effective mean photovoltaic output follows the hierarchy: High-Irradiance Year > Normal-Irradiance Year > Low-Irradiance Year. Additionally, Qinghai's solar irradiance exhibits distinct seasonal patterns: higher levels from January to May and October to December and lower levels from June to September. Notably, the Low-Irradiance Year demonstrates the lowest average irradiance intensity in June, with an output influence factor of 0.367.

Analysis of historical wind and solar output time-series data reveals distinct seasonal patterns in Qinghai's renewable resources, which align with the hydrological wet and dry seasons. This correlation can be explained as follows: during the dry season, lower frequencies of cloudy/rainy weather enable higher wind and solar output levels. Conversely, during the wet season, increased rainfall frequency suppresses renewable generation, resulting in reduced output levels.

2.2. Combined unit model for power systems

To evaluate the potential contribution of SPS to flexibility in power systems with high penetration of renewable energy, this section develops a regional power system optimal dispatch model incorporating SPS. By leveraging the output influence factors of typical wind-solar scenarios derived from the k-means++ clustering in the previous section, the model simulates unit operational characteristics under different renewable energy scenarios.

2.2.1. Objective function

(1) The objective function of the power system optimization dispatch model aims to minimize total operating costs over the scheduling horizon, including start-up and shut-down costs, fixed costs, carbon emission costs, ramping costs, loss of load cost, power curtailment costs, transmission cost. The objective is formulated as:

$$C_{total} = \min \sum_{u,i} \left(C_{u,i}^{start} + C_{u,i}^{off} + C_{u,i}^{fix} + C_{u,i}^{ramp+} + C_{u,i}^{ramp-} + C_{u,i}^{curtailment} + C_{l,i}^{trans} + C_{n,i}^{loss} + C_{n,i}^{carbon} \right) \tag{5}$$

Where subscript u denotes all generating units; subscript i represents hourly time steps; n refers to the unit's location; C_{total} is the total operating cost; $C_{u,i}^{start}$ and $C_{u,i}^{off}$ are the start-up and shut-down costs of the unit; $C_{u,i}^{fix}$ is the fixed costs of running the unit u in all time periods i; $C_{u,i}^{ramp+}$ and $C_{u,i}^{ramp-}$ are respectively represent the ramp up cost and ramp down cost of unit u in all time periods i; $C_{u,i}^{curtailment}$ is the power curtailment cost; $C_{l,i}^{trans}$ is the transmission cost directly through the lines; $C_{n,i}^{loss}$ is the l*oss of load cost,* $C_{n,i}^{\text{carbon}}$ is the carbon emission cost.

(1) **Start-up and shut-down costs**

The start-up and shutdown costs of the unit mainly come from the fuel cost generated by starting or shutting down. The start-up cost and shutdown cost can be expressed as:

$$\begin{aligned} C_{u,i}^{start} &= c^{start} x_{u,i} \\ C_{u,i}^{off} &= c^{off} x_{u,i} \end{aligned} \tag{6}$$

Where $x_{u,i}$ is the binary state variable for each unit at each time step, where 0 indicates the unit is offline and 1 signifies it is online; c^{start} and c^{off} are the startup or shutdown costs generated by unit u at one time.

(2) **Fixed costs**

Fixed costs represent operation and maintenance costs and other costs associated for operating the unit and it can be described as:

$$C_{u,i}^{fix} = c^{fix} x_{u,i} p_{u,i} \tag{7}$$

Where c^{fix} is the fixed cost incurred during operation of the unit in the on state $p_{u,i}$ are the amount of electricity generated by the unit.

(3) **Carbon emission costs**

The carbon emission cost was calculated using the carbon emission factor method [39]. For the 300 MW conventional coal-fired units in Qinghai Province, the grid emission factor was 0.979 tCO_2/MWh and it can be described as:

$$C_{n,i}^{carbon} = c^{carbon} p_{n,i} \tag{8}$$

where *ccarbon* is carbon emission cost coefficient for power generation.

(4) **Ramping costs**

Ramping cost mainly refers to the cost generated by unit output adjustment, which can be expressed as:

$$\begin{aligned} C_{u,i}^{ramp+} &= c^{ramp+}(P_{u,i} - P_{u,i-1}) \\ C_{u,i}^{ramp-} &= c^{ramp-}(P_{u,i-1} - P_{u,i}) \end{aligned} \tag{9}$$

where c^{ramp+} and c^{ramp-} represent ramping up and ramping down costs, respectively.

(5) **Loss of Load cost**

This cost can be used to assess the economic losses caused by power shortages, in Qinghai Province is 4,051 CNY/MWh [40]. It was calculated based on the province's GDP in 2020 (300.592 billion CNY) and annual electricity consumption (74.2 billion kwh) using the power generation ratio method.

$$C_{n,i}^{loss} = c^{loss} d_{n,i} \tag{10}$$

Where c^{loss} is the cost corresponding to a loss of 1MWh load in the area n; $d_{n,i}$ is the amount of lost load.

(6) **Power curtailment costs**

Power curtailment cost includes hydro, wind and solar curtailment cost [41]. Hydro curtailment cost sets the hydro spillage in Qinghai Province at 800 CNY/MWh. The cost of wind and solar power curtailment is calculated 0.3196 CNY/kWh.

$$\begin{aligned} C_{n,i}^{curtail(h)} &= c^{h} p_{n,i}^{h} \\ C_{n,i}^{curtail(s)} &= c^{s} p_{n,i}^{s} \\ C_{n,i}^{curtail(w)} &= c^{w} p_{n,i}^{w} \end{aligned} \tag{11}$$

Where c^h,c^s and c^w are the costs per MWh of hydro, solar and wind curtailment in area n, respectively;$p_{n,i}^h$,$p_{n,i}^s$ and $p_{n,i}^w$ are the total amounts of curtailed hydro, solar and wind energy in area n, respectively.

(7) **Transmission cost**

The transmission cost is based on the regional power grid transmission price standards announced by the National Development and Reform Commission in 2020 [42]. In Qinghai Province, the transmission cost electricity price is 0.02 CNY/kWh.

$$C_{l,i}^{trans} = c^{trans} p_{l,i} \tag{12}$$

where c^{trans} is the cost per MWh of electricity transmitted through transmission line l, and $p_{l,i}$ is the amount of electricity transmitted.

2.2.2. Power balance constraint

The power balance equation ensures supply–demand equilibrium at each time step, requiring that the total active power from generating units plus the net external power exchange equals the load demand plus pumping power consumption of pumped storage minus the system flexibility deficit power at any time step.

$$\sum_i P_{u,i} + \sum_i Pf_{u,i} = D_i + \sum_P S_i - CUR_i \tag{13}$$

where D_i is total load demand at time step *i*; S_i indicates pumping power consumption of energy storage systems at time step *i*. Where D_i is total load demand at time step i; S_i indicates pumping power consumption of energy storage systems at time step i. CUR_i represents the system flexibility deficit at time step *i*, encompassing both unserved load (load shedding) and curtailed renewable energy (wind, solar, or run-of-river hydropower curtailment) when the available generation cannot fully meet demand or absorb surplus.

2.2.3. Operational constraints of seasonal pumped storage

Unlike CPS, SPS is uniquely characterized by its ability to operate across both short-term and long-term cycles. To capture this distinction, the model explicitly incorporates the following operational features of SPS:

(1) **Power output constraints**

For SPS, the power output must exceed the minimum stable output level and remain below the maximum rated output level of the unit. When the unit is in the offline state, its output must be zero [43,44].

$$P_{p,i}^{\min} \cdot g_p \cdot c_{p,i} \le P_{p,i} \le P_{p,i}^{\max} \cdot g_p \cdot c_{p,i} \tag{14}$$

where subscript *p* denotes pumped storage units; $P_{p,i}^{min}$ and $P_{p,i}^{max}$, respectively, represent the minimum and maximum output power percentages of SPS; g_p denotes the rated output power of the SPS.

To ensure power system stability and reliability, the power output of SPS units is constrained by the previous time step's output level and the upward/downward ramping rates [45,46], formulated as follows:

$$P_{p,i-1} - RD_p \cdot g_p \cdot 60 \le P_{p,i} \le P_{p,i-1} + RU_p \cdot g_p \cdot 60 \tag{15}$$

where RD_p and RU_p, respectively, denote the downward ramping rate and upward ramping rate of SPS.

(2) **Pumping operation constraints**

The pumping operation of SPS must also operate within stable limits. Specifically, the pumping power at any time step must exceed the minimum pumping level and remain below the maximum rated output level of the unit. This is mathematically formulated as follows:

$$Q_{p,i}^{\min} \cdot p_p \cdot c_{p,i} \le S_{p,i} \le Q_{p,i}^{\max} \cdot p_p \cdot c_{p,i} \tag{16}$$

where $Q_{P,i}^{min}$ and $Q_{P,i}^{max}$, respectively, denote minimum and maximum pumping power percentages of SPS. P_p is the rated pumping power of the SPS.

Similarly, the pumping operation of SPS is constrained by the pumping power from the previous time step, upward ramping rate, and downward ramping rate, formulated as follows:

$$S_{p,i-1} - RD_P \cdot p_p \cdot 60 \le S_{p,i} \le S_{p,i-1} + RU_p \cdot p_p \cdot 60 \tag{17}$$

(3) **Generation-pumping switching constraints**

The SPS units selected in this study are fixed-speed units, and the operational mode transition constraints between pumping and generation at any time step can be formulated as follows:

$$S_{p,i} \le p_{p,i} \cdot (1 - c_{p,i}) \tag{18}$$

(4) **Reservoir storage capacity constraints**

Reservoir storage capacity serves as the primary constraint on the

energy storage level of SPS, meaning the actual storage level at any time step must remain between the maximum and minimum available capacities. This is formulated as:

$$RC_{p,i}^{\min} \le RC_{p,i} \le RC_{p,i}^{\max} \tag{19}$$

where $RC_{p,i}$ is the actual storage level within a single time step for pumped storage; $RC_{p,i}^{min}$ and $RC_{p,i}^{max}$, respectively, denote the minimum and maximum available capacities of the reservoir.

The power generation and pumping operations of SPS must satisfy the reservoir water balance constraint. Specifically, the storage capacity of the reservoir at time step i is equal to the sum of the inflow volume, pumped water volume and the previous time step's storage capacity minus the generation discharge, spillage, and mandatory release during time step i. This is formulated as:

$$RC_{p,i} = RC_{p,i-1} + S_{p,i}\eta_p + Q_{p,i}^{in} - Q_{p,i}^{out} - Q_{p,i}^{sp} - P_{p,i}/\eta_g \tag{20}$$

where η_p *and* η_g, respectively, denote pumping efficiency and generation efficiency; $Q_{s,p}^{\text{in}}$ represents inflow volume; $Q_{s,p}^{\text{out}}$ indicates reservoir spillage; $Q_{s,p}^{\text{sp}}$ corresponds to discharge flow.

The electricity generation of SPS units is constrained by the available storage capacity of the upper reservoir. This is formulated as:

$$P_{p,i}/\eta_g + Q_{p,i} + Q_{p,i}^{sp} - Q_{s,p}^{in} \le RC_{p,i} \tag{21}$$

The charging capacity of SPS is constrained by the reservoir's available storage capacity. This is formulated as:

$$S_{p,i}\cdot\eta_p - Q_{p,i}^{out} - Q_{p,i}^{sp} + Q_{p,i}^{in} \le Ec_{s,i} - RC_{p,i} \tag{22}$$

Transmission capacity constraints, unit operational constraints, reservoir operational dispatch constraints, and others are defined in detail in reference [47,48]. These constraints are not elaborated here.

(5) **Transmission constraints**

To accurately model the inter-regional power exchanges and ensure grid feasibility, the optimization framework incorporates transmission capacity constraints. The power flow on each transmission corridor l connecting different nodes is constrained by its physical capacity limits:

$$F_{L,\min} \le F_{L,i} \le F_{L,\max} \tag{23}$$

where $F_{L,i}$ is the power flow on corridor l at time i, and $F_{L,\min}$ and $F_{L,\max}$ are the minimum and maximum transmission capacities of the corridor, respectively. Together, these constraints ensure that all scheduled power dispatches are consistent with the physical limitations of the transmission network.

2.2.4. Model solving

The model is formulated as a Mixed-Integer Linear Programming (MILP) problem and solved in GAMS. MILP outperforms Linear Programming (LP) by precisely handling integer variables (e.g., unit commitment, reservoir capacity) while optimizing continuous variables like power output and water levels. The GAMS solver leverages advanced algorithms such as branch-and-bound to ensure accuracy while significantly enhancing computational efficiency and convergence.

2.2.5. Data sources

(1) Wind and Solar Power Output Series

The uncertainty of wind and solar power output poses a significant challenge to power system dispatch. In power system modeling, the available power output from these sources is typically represented by the Eq. (24):

$$g_{a,i}^{k} = g_a A_{a,i} \tag{24}$$

where $g_{a,i}^{k}$ is the actual power output of the wind and solar units; $A_{a,i}$ is the effective output coefficient, which ranges from 0 to 1.

(2) Calculation of the PV and wind availability factors

Availability factors for PV and wind power are calculated using analogous methodologies. The process begins by inputting the geographic coordinates of the selected site into specialized platforms: *https://globalsolaratlas.info/map* for PV and *https://www.renewables.ninja* for wind power. Key parameters are specified, including the technology type and mounting setup for PV, and the hub height and turbine model for wind power. These platforms then simulate the hourly theoretical power output based on integrated models and reanalysis meteorological data. The availability factor time series is subsequently obtained by normalizing this simulated power output with the respective rated capacity. The detailed model algorithms are referenced from [49,50].

(3) Processing of runoff data

The hydropower runoff time series for the Qinghai Plateau region was constructed by integrating the atlite toolkit [50], developed by Prof. Fabian Hofmann, with the hydrological modeling approach proposed [51].First, the watershed units within Qinghai Province were extracted from the global Hydrobasins dataset. These units were then topologically delineated using the Pfafstetter coding system. Secondly, the topographical data and grid runoff data for the study area were retrieved from the ERA5 global reanalysis dataset. The study area itself is delineated by the administrative boundaries of Qinghai Province. Then, for each of the major reservoirs in Qinghai Province, its upstream catchment area was delineated using the atlite toolkit based on its geographical coordinates. Subsequently, the gridded runoff data within each catchment were converted into a time series of reservoir inflow.The reservoir inflow is converted into power generation by eq. 25, which incorporates the design head of each hydropower station in Qinghai Province [52].Finally, the total daily streamflow was uniformly distributed over 24 h to generate the hourly streamflow time series.

$$Z_{\max}(kW) = 9.81\rho_w V h_t \eta_t / 3.6 \times 10^6 \tag{25}$$

where,ρ_w is the water density, with a value of approximately 1000 kg/m^3; h_t is the generating head of the turbine;η_t is the energy conversion efficiency (dimensionless), which typically ranges from 0.8 to 0.9 for conventional hydropower plants and 0.75 to 0.77 for pumped-storage hydropower plants.

(4) Load demand data

The load demand time series serves as a fundamental input for power system operation modeling. This study synthesizes a time series, leveraging the typical provincial grid load profiles [53] and annual energy consumption data [54]. The calculation is given by Eq. (26).

$$E_{\text{i}} = (E_{t,i} - E_{t,\min})/(E_{t,\max} - E_{t,\min})(D_{\max} - D_{\min}) + D_{\min} \tag{26}$$

where E_i is the hourly load demand; $E_{t,i}$ is the hourly typical daily load profile; $E_{t,\max}$ and $E_{t,\min}$ represent the max and min daily load factors, respectively; and D_{max} and D_{min} denote the daily max and min loads, which are derived from the published annual daily peak and valley load curves.

2.3. Comprehensive benefit evaluation model

To quantify the comprehensive benefits of SPS, this section establishes a multi-criteria assessment framework. Firstly, key indicators are

selected across technical, economic, and operational stability criteria, as shown in Fig. 6. Secondly, an integrated weighting approach combining the entropy-weighted and the coefficient of variation methods is applied.

2.3.1. Technical-economic-stability indicator system

(1) System technical indicators

Curtailment Rate (CTR): defined as the ratio of energy curtailment (from wind, solar, and run-of-river hydropower) to the total available renewable energy generation during the studied period.

$$CTR = \sum_{i=0} Curtailment_i / \sum_{i=0} (Pt_i^{wind} + Pt_i^{pv} + Pt_i^{stable}) \times 100\% \quad (27)$$

where $Curtailment_i$ represents the system curtailment at any time step i.

Net Load Peak-Valley Difference (NPD): Net load is defined as the system's total electricity demand minus the power generated from variable renewable energy sources. The Net Load Peak-Valley Difference (NPD) is subsequently defined as the absolute difference between the maximum and minimum net load values observed within a given study period. This metric is crucial for flexibility assessment, as it quantitatively reflects the variability of the net load. A larger NPD signifies more pronounced peak-to-trough fluctuations, which in turn necessitates that the power system possess enhanced flexible regulation capacity to maintain balance during these steep ramping cycles [55].

$$\begin{aligned} L_i^{net} &= D_i - Pt_i^{wind} - Pt_i^{pv} \\ NPD &= \max(L_i^{net}) - \min(L_i^{net}) \end{aligned} \quad (28)$$

where L_i^{net} denotes the net load at any time step i.

Carbon Emissions (CEM): originates from coal-fired power generation and exhibits a positive correlation with its electricity output[51]. This is formulated as:

$$PF_{f,i} = P_{f,i}^{THE} Tp_f \quad (29)$$

where f denotes thermal power units; $PF_{f,i}$ represents the CEM at time step i; Tp_f is the carbon emission coefficient of thermal power.

Flexibility Adequacy Rate (FAR): measures the share of time a power system maintains sufficient flexibility to avoid shortages (load shedding) or renewable energy waste (no curtailment). This is formulated as:

$$FAR = \frac{I_{cy}}{I} \times 100\% \quad (30)$$

where *flex* denotes the system FAR; I_{cy} represents the flexibility adequacy duration; I indicates the total system operating time.

Flexibility Deficiency Rate(FDR): it measures the severity of system flexibility inadequacy, calculated as the ratio of flexibility deficit to net load. The FDR directly indicates system reliability risks: higher values imply a greater probability of load shedding or renewable curtailment, while lower values demonstrate stronger operational flexibility. This is formulated as:

$$FDR = \sum_{i}^{I} \left| P_i^{dem} - P_i^{\sup} \right| / \sum_{i}^{I} L_i^{Net} \times 100\% \quad (31)$$

where P_i^{dem} and P_i^{sup}, respectively, denote the system flexibility demand and system flexibility supply at time step i.

(2) System economic indicators

Capital Cost (CAP): both SPS and CPS require the construction of upper and lower reservoirs, pressure pipelines, and pump-turbine units. Based on existing studies [56], the CAP for SPS plants is 25.165 million CNY per megawatt (MW), while CPS plants incur a lower CAP of 18.967 million CNY per MW[19].

System Operating Cost (OPC): includes unit start-up/shutdown costs, fixed costs, variable costs, carbon emission costs, ramping costs,

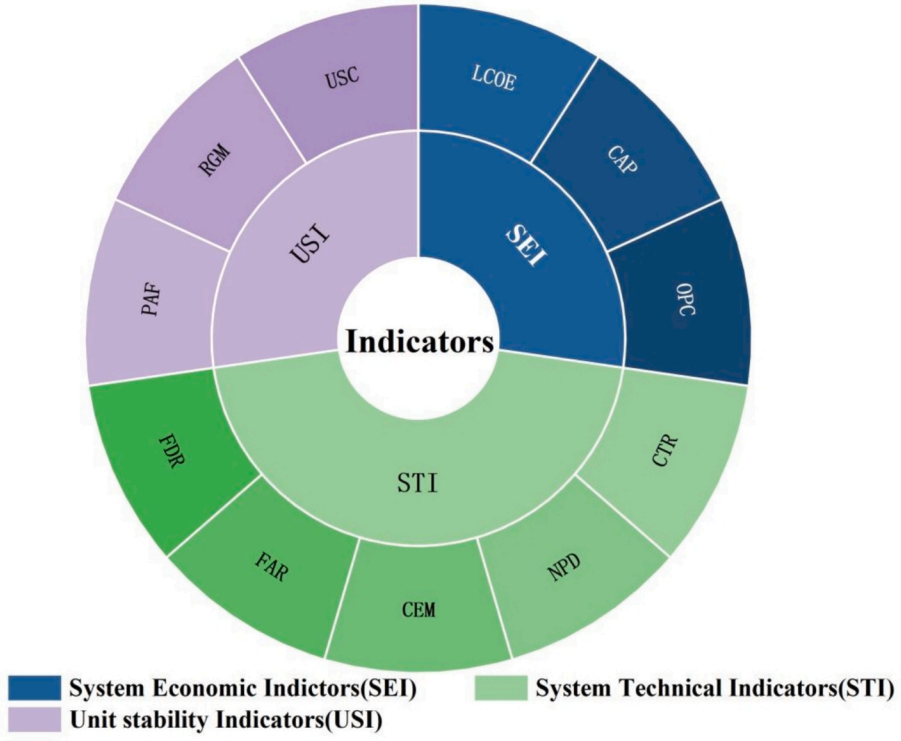


Fig. 6. Seasonal pumping reserve evaluation index system.

transmission costs, and load shedding costs. The methodology for calculating these costs is outlined in Subsection 2.3.1, and the parameters associated with operating costs for different generation categories are summarized in Table. 1.

Levelized Cost of Electricity (LCOE): This metric represents the ratio of the total costs incurred over the project's operational life to the total electricity generated. as shown in the following formula:

$$LCOE = \frac{C_{CAP} + \sum_{n=1}^{N} C_{OM}/(1+r)^n}{\sum_{n=1}^{N} E_n/(1+r)^n} \tag{32}$$

where C_{OM} is operation and maintenance costs;E_n represents the total electricity output over the technical lifetime; N denotes the technical lifespan (in years); r refers to the standard discount rate.

(3) Unit stability indicators

Power Adjustment Frequency (PAF): it counts the annual occurrences of output power changes in pumped storage units. A change is recorded when the power output difference between consecutive hours is non-zero. This is formulated as:

$$PAF = \sum_{i=1}^{8760} \mathbb{I}(\Delta P_i \neq 0), \Delta P = P_{i+1} - P_i \tag{33}$$

Regulation Mileage (RGM): it is defined as the mean absolute power adjustment per operational change, quantifying the typical magnitude of output variations in flexible generation units, as expressed in Eq. (36):

$$RGM = \frac{\sum_{i=1}^{i} |\Delta P_i|}{PAF} \tag{34}$$

Unit Start Frequency (USF): it quantifies the annual start-up counts of pumped storage units in both turbine mode and pumping modes. Variations in operational strategies across different technologies lead to distinct start-up frequencies for each model [57]. This is formulated as:

$$USF = \begin{cases} turbine : \sum_{i=1}^{8760} \mathbb{I}(P_t = 0 \cap P_{t+1} > 0) \\ pump : \sum_{i=1}^{8760} \mathbb{I}(P_t = 0 \cap P_{t+1} < 0) \end{cases} \tag{35}$$

2.3.2. Entropy-weighted variation coefficient integrated assessment model

In the field of multi-criteria decision analysis (MCDA), several methods such as the Analytic Hierarchy Process (AHP) and the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) are widely employed. However, methods like AHP can introduce bias, particularly in nascent research areas like the holistic valuation of SPS where expert consensus is still forming [53,54].

To overcome this limitation, the trend in energy systems assessment has been moving towards objective, data-driven weighting methods that minimize human intervention [55].This section proposes an integrated evaluation method combining entropy-weighted and variation coefficients. By leveraging the strengths of both approaches, this method incorporates the information entropy characteristics of the data while accounting for the variability of indicators, thereby enhancing the rationality and reliability of weight allocation, and mitigating the subjectivity of traditional approaches. The specific steps are as follows:

(1) Dimensionless normalization of indicators

To improve evaluation accuracy, Z-score standardization is applied to normalize all indicators and eliminate dimensional effects, as expressed in Eq. (38):

$$y_{i,j} = \frac{x_{i,j} - mean(x_j)}{std(x_j)} \tag{36}$$

where $X_{i,j}$ is raw indicator data; $y_{i,j}$ is standardized data after normalization.

(2) Entropy-weight method

The entropy-weighted method prioritizes objective weighting, where the assigned weights are determined by the dispersion of the data, effectively mitigating subjective bias [56]. The weighting coefficients are calculated.

$$e_j = -\sum_{i=1}^{n} y_{ij} \frac{\ln y_{ij}}{\ln n} \tag{37}$$

where e_j denotes the entropy value of the *j-th* indicator; n represents the number of scenarios.

$$sw_j = \frac{1 - e_j}{\sum_{i=1}^{m}(1 - e_j)} \tag{38}$$

where sw_j denotes the indicator weight assigned by the entropy-weighted method; m represents the total number of evaluation indicators.

(3) Coefficient of variation method

The coefficient of variation method assigns weights based on the variability of indicators, measuring the deviation of evaluated entities from the average indicator level. This approach enables dynamic weighting of indicators for assessed entities [58].

$$\beta_i = \frac{std(x_i)}{mean(x_i)} \tag{39}$$

where β_i denotes the coefficient of variation for the *i-th* indicator; $std(x_i)$ and $mean(x_i)$, respectively, represent the standard deviation and arithmetic mean of the indicator.

$$bw_j = \frac{\beta_i}{\sum_{i=1}^{m}\beta_i} \tag{40}$$

where bw_j denotes the indicator weight assigned by the coefficient of variation method; m represents the total number of evaluation indicators.

Comprehensive weighting is performed using the Lagrange multiplier method [59] to integrate the results from both weighting approaches, as formalized in Equation (41):

$$zw_j = \frac{(sw_j \cdot bw_j)^{1/2}}{\sum_{i=1}^{m}(sw_j \cdot bw_j)^{1/2}} \tag{41}$$

where zw_j denotes the integrated weighting coefficient derived from the combined entropy-weighted method and coefficient of variation method.

Table 1
The operating costs of different generation technologies (unit: 10,000 CNY/MW).

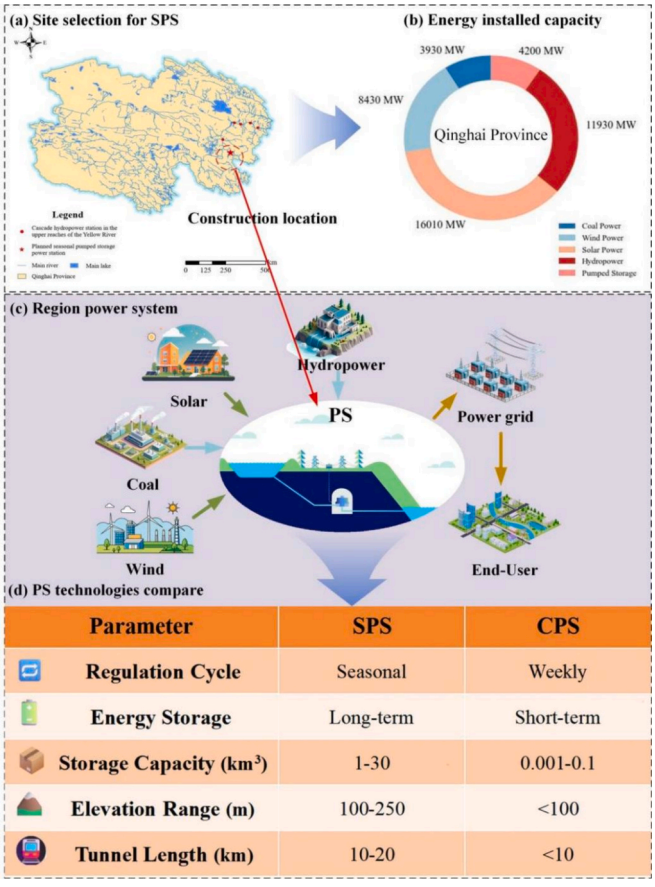
Type	Start-up costs	Variable costs	Fixed costs	Peak-Shaving costs
Hydropower	0	0	25.93	0
Thermal	0.11	0.0274	6.2	0.0041
Solar	0	0	6.65	0
Wind	0	0	14.6	0
Pumped storage	0	0	14.92	0

3. Case study

3.1. Case description

China's Qinghai Province (36.56°N, 101.74°E) is characterized by a plateau continental climate and abundant renewable energy resources. Firstly, Qinghai exemplifies the acute challenges of a renewable-dominated power system, with wind and solar installations constituting over 60 % of its total capacity, with wind power at 8,430 MW and solar at 16,010 MW. Fig. 7b illustrates the installed energy capacity in Qinghai Province, while Fig. 7c presents the composition of its power system. And evidenced by a renewable energy curtailment rate reaching 17 % in 2021, signifying a substantial deficit in system flexibility. Secondly,the region exhibits pronounced seasonal disparities in both renewable generation and load demand. Therefore, Qinghai can investigate and quantify the unique value proposition of Seasonal Pumped Storage (SPS) in addressing mid-to-long-term flexibility deficits.

To address the lack of flexibility in the power system, Qinghai Province has initiated plans to deploy multiple pumped storage plants along the Yellow River basin, with a total installed capacity of 4,200 MW, aiming to enhance grid regulation capacity [57]. Based on geological and siting criteria, the proposed SPS plant is assumed to be located at the red star marker in Fig. 7(a), with a planned installed capacity of 4,200 MW. The upper reservoir is sited along a primary tributary of the Yellow River, while the lower reservoir utilizes the existing Longyangxia Hydropower Station Reservoir. These two reservoirs are connected via a 21-kilometer pressure pipeline, forming a natural hydraulic gradient. This unique geographical configuration provides optimal hydrological conditions for SPS operations.



Parameter	SPS	CPS
Regulation Cycle	Seasonal	Weekly
Energy Storage	Long-term	Short-term
Storage Capacity (km^3)	1-30	0.001-0.1
Elevation Range (m)	100-250	<100
Tunnel Length (km)	10-20	<10

Fig. 7. Overview of geographical regions of Qinghai province.

3.2. Scenario setting

To rigorously assess and compare the comprehensive performance of Seasonal Pumped Storage (SPS) and Conventional Pumped Storage (CPS) amidst the inherent stochastic fluctuations of wind and solar power, this section meticulously constructs a set of scenario variables. These variables are systematically defined across three critical dimensions, culminating in 18 distinct scenario combinations, which are comprehensively summarized in Table 2. This multi-dimensional scenario framework is designed to capture a wide range of operational conditions and renewable energy availabilities, thereby providing a robust basis for our comparative analysis.

(i) **Pumped storage Technology Type**: We distinguish between Conventional Pumped Storage (CPS) systems, which are typically designed for daily or intra-day regulation cycles, and Seasonal Pumped Storage (SPS) systems. SPS, in contrast, is characterized by its significantly larger reservoir capacities and inherent ability to perform seasonal or multi-seasonal energy regulation, thus offering distinct mid-to-long-term flexibility. Key operational and design parameter differences between these two technologies, particularly regarding energy storage capacity and discharge duration, are elaborated and presented in Fig. 7(d) [60]. For modeling purposes, CPS is denoted as 'PS' and SPS as 'JP'.

(ii) **Wind power Generation Conditions:** Acknowledging the pronounced variability of wind resources, we leveraged the advanced k-means++ clustering algorithm (as detailed in Section 2.1) on extensive historical wind data. This enabled us to robustly identify and classify typical annual wind patterns into three representative sub-scenarios: Low-Wind Year (LWY, denoted as 'a'), characterized by consistently lower average wind speeds; Normal-Wind Year (NWY, denoted as 'b'), representing average wind conditions; and High-Wind Year (HWY, denoted as 'c'), reflecting periods of sustained high wind speeds and energy output.

(iii) **Solar output levels:** Irradiance Levels: Similar to wind power, the intermittent and geographically dependent nature of solar power necessitates a robust characterization. Utilizing the same k-means++ clustering methodology (further elaborated in Section 2.1), we identified and categorized annual solar irradiance patterns into three representative sub-scenarios: Low-Irradiance Year (LIY, denoted as 'A'), indicating periods of reduced solar radiation; Normal-Irradiance Year (NIY, denoted as 'B'), representing average irradiance conditions; and High-Irradiance Year

Table 2
Scenarios division results.

Scenarios	Types of pumped storage	Wind power characteristics	Solar power characteristics
JPaA	SPS	LWY	LIY
JPaB	SPS	LWY	NIY
JPaC	SPS	LWY	HIY
JPbA	SPS	NWY	LIY
JPbB	SPS	NWY	NIY
JPbC	SPS	NWY	HIY
JPcA	SPS	HWY	LIY
JPcB	SPS	HWY	NIY
JPcC	SPS	HWY	HIY
PSaA	PS	LWY	LIY
PSaB	PS	LWY	NIY
PSaC	PS	LWY	HIY
PSbA	PS	NWY	LIY
PSbB	PS	NWY	NIY
PSbC	PS	NWY	HIY
PScA	PS	HWY	LIY
PScB	PS	HWY	NIY
PScC	PS	HWY	HIY

(HIY, denoted as 'C'), signifying periods of optimal and sustained high solar irradiance.

4. Results

4.1. Technical performance analysis

To visually compare the impact of SPS and CPS on system CTR under varying wind-solar conditions, the CTR across all 18 scenarios is presented as line charts in Fig. 8.

SPS demonstrates notable technical advantages. In extreme scenarios with high wind and irradiance, SPS (JPcC and PScC scenarios) achieved an August CTR of 4.35 %-just 24 % of the rate observed in CPS (18.14 %). Additionally, SPS's annual CTR (5.47 %) was significantly lower than that of CPS (12.3 %). This advantage stems primarily from the cross-seasonal energy storage capability of SPS, which stores surplus renewable electricity generated in summer for winter use. Under mild scenarios in Low-Wind and Low-Irradiance, the difference narrows to SPS (JPaA scenario) 1.93 % and CPS (PSaA scenario) 1.80 %, demonstrating comparable regulation effectiveness between the two technologies under low renewable output conditions. Regarding seasonal applicability, SPS demonstrates particularly pronounced advantages during the summer months (June-August). August data reveals that in scenarios with moderate-to-high wind-solar output levels, SPS achieves systematically lower CTR compared to CPS. Notably, under high wind-solar output conditions, CPS exhibits sharp curtailment spikes (e.g., PScC increased 32 % from July to August), whereas SPS maintains gradual increases (e.g., JPcC: 12 %). This contrast demonstrates SPS's superior regulation stability.

These findings demonstrate that SPS effectively mitigates curtailment challenges in high-penetration renewable energy systems through its unique multi-temporal flexibility regulation capabilities, particularly during summer months with elevated wind-solar generation. However, SPS's regulatory advantages diminish significantly in seasons with lower renewable energy output.

To visually compare the impacts of SPS and CPS on net load fluctuations across diverse scenarios, Fig. 9 presents the daily average NPD for each month under all 18 scenarios.

Throughout the year, NPD exhibits a characteristic summer-low and winter-high pattern. Specifically, from June to September, the peak-to-valley differences remain relatively low across all scenarios (SPS: 8,000–11,000 MW; CPS: 8,000–12,000 MW). During winter load peaks (November to February), these differences increase significantly (SPS: 11,000–13,000 MW; CPS: 11,000–13,500 MW). This seasonal pattern aligns closely with the cyclical variations in load demand and renewable energy output observed in Northwest China. Regarding flexibility regulation capability, SPS demonstrates superior performance. Under Low-Wind and Low-Irradiance scenarios, SPS system achieves an annual average peak-to-valley difference of 12,156 MW, representing a 1.5 % reduction compared to CPS (12,345 MW). This advantage amplifies with increased wind-solar output. In High-Wind and High-Irradiance, SPS (JPcC scenario) shows a 10.6 % reduction in annual average peak-to-valley difference (11,234 MW) relative to the comparable CPS (PScC scenario: 12,567 MW). Notably, during the extreme cold periods of January-February, SPS demonstrates significantly superior regulation performance compared to CPS. Taking SPS (JPcA scenario) and CPS (PScA scenario) as examples, the peak-valley differences in January were 11,425 MW and 11,424 MW, while in February they were 11,208 MW and 10,824 MW. This represents a 3.5 % reduction in peak-valley difference for SPS relative to CPS.

The variability of wind and solar power generation significantly impacts system carbon emissions. To quantitatively assess the emission differences between SPS and CPS across scenarios, Fig. 10 presents a carbon emission stacked chart analysis.

As shown in Fig. 10, SPS exhibits significantly lower CEM than CPS across all 18 scenarios. In the typical High-Wind and High-Irradiance, SPS (JPcC scenario) achieves zero annual CEM, whereas CPS (PScC scenario) emits 2.72×10^9 tons–a striking disparity. Even under Low-Wind and Low-Irradiance, CPS demonstrates substantially higher emissions ($2.02–2.90 \times 10^9$) compared to SPS ($0–1.72 \times 10^7$), demonstrating SPS's effectiveness in reducing CEM even during periods of low renewable energy output. The results from SPS (JPcA, JPcB, and JPcC scenarios) demonstrate it achieves zero annual CEM under High-Wind conditions, indicating its ability to fully replace thermal power generation for grid regulation in these scenarios. For CPS in PScA, PScB, and PScC scenarios, CEM primarily concentrates between April and July, with no significant increase observed in other months. This analysis reveals that SPS exhibits stronger carbon emission-curbing effects under higher wind conditions.

To assess system flexibility across different generation scenarios, this study utilizes the FAR to measure flexibility sufficiency and the FDR to quantify insufficiency. Fig. 11 presents 3D bar charts comparing the flexibility performance of SPS and CPS across 18 scenarios: (a)-(c) show FAR results, while (d)-(f) display FDR outcomes.

According to the FAR results in Fig. 11, SPS demonstrates superior overall performance compared to the CPS across all operational scenarios. Under Low-Wind and Low-Irradiance, SPS's (JPaA scenario) FAR achieves its peak at 86.03 %. This indicator gradually decreases to 76.55 % in JPcC with higher wind and irradiance. In High-Wind scenarios (c-series), the FDR for SPS (JPcA scenario) in February (0.192) shows a 735 % increase compared to SPS (JPaA scenario) (0.023), Disparity in FAR between the two pumped storage technologies is concentrated during the high-flow season (July-October). For example, in September, SPS (JPcC scenario) achieves a rate of 0.649, while CPS (PScC scenario)

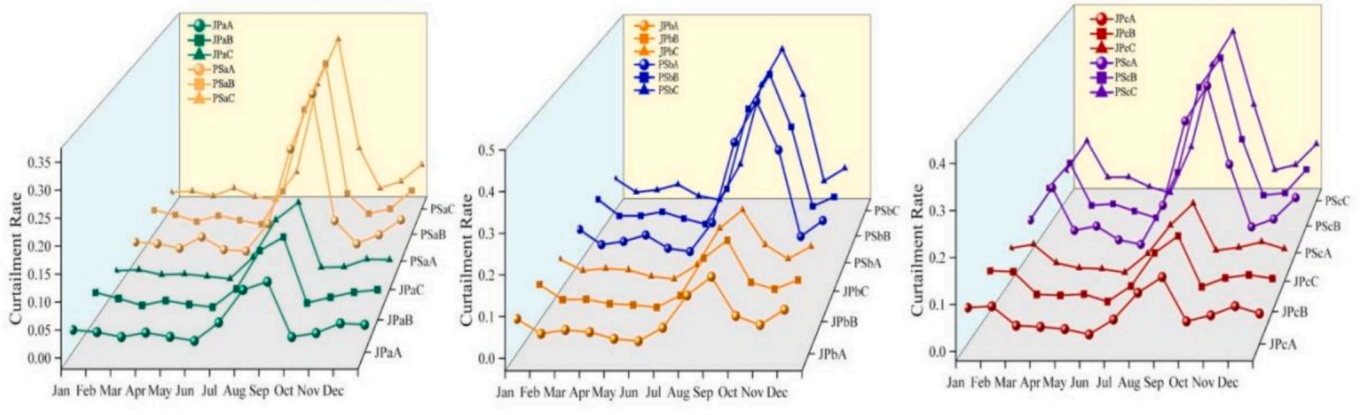


Fig. 8. Curtailment Rates of 18 scenarios for each month.

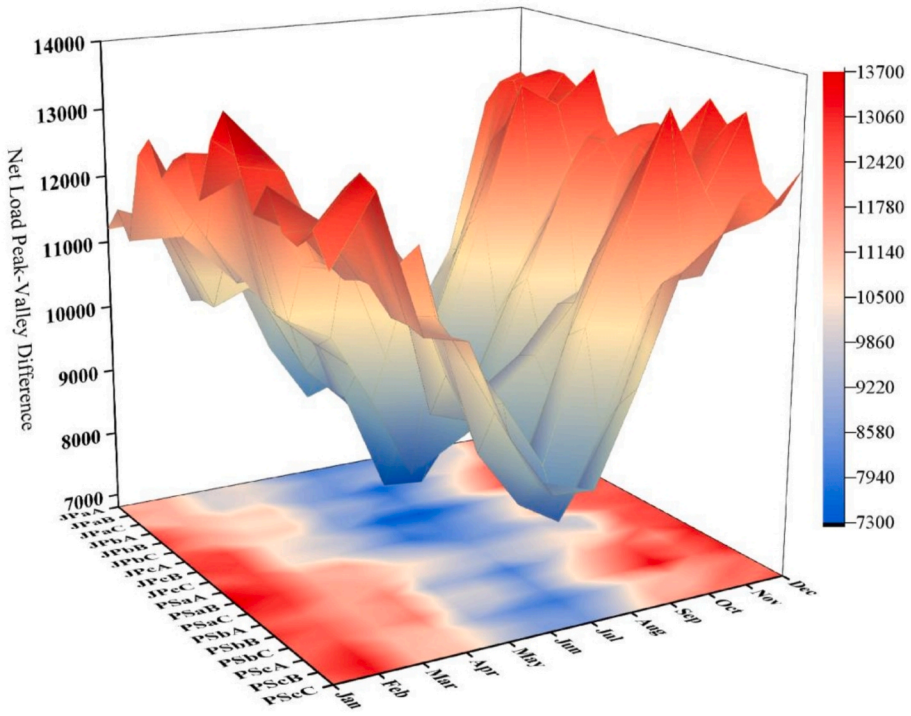


Fig. 9. Heatmap of average net load peak-to-valley differences for each month across.

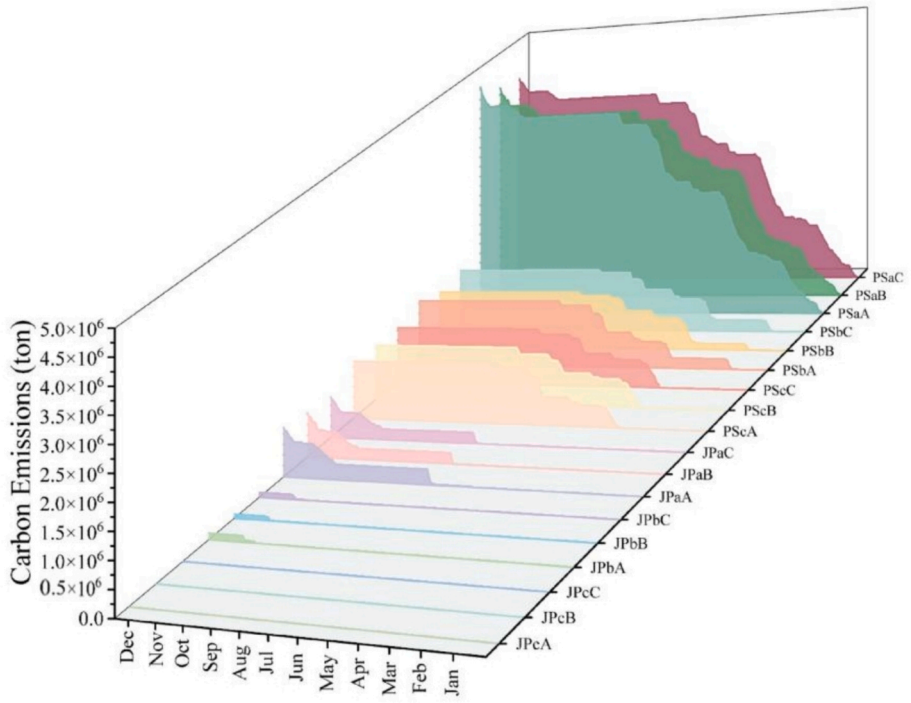


Fig. 10. Carbon emission of 18 scenarios for each month.

reaches only 0.271–139.5 % improvement for SPS. Conversely, during the low-flow season (January-March), CPS slightly outperforms SPS: in February, CPS (PSaC scenario) registers 0.863 compared to SPS (JPaC scenario) at 0.839. During high-flow seasons, SPS exhibits significantly higher FAR than CPS. Conversely, CPS slightly outperforms SPS in low-flow seasons. This disparity arises because SPS prioritizes continuous power generation during dry periods, which constrains its pumping operations, whereas CPS operates under daily regulation cycles, enabling frequent operational mode switching to better support grid flexibility demands. During high-flow seasons, SPS outperforms CPS due to its larger regulation storage capacity, enabling sustained pumping operations that significantly enhance system FAR. Consequently, SPS exhibits superior flexibility adequacy performance compared to CPS.

The FDR in Fig. 11 demonstrates that SPS consistently outperforms CPS, with this advantage amplifying under higher wind-solar intensities. The annual average deficiency rate reaches its minimum value of 0.049

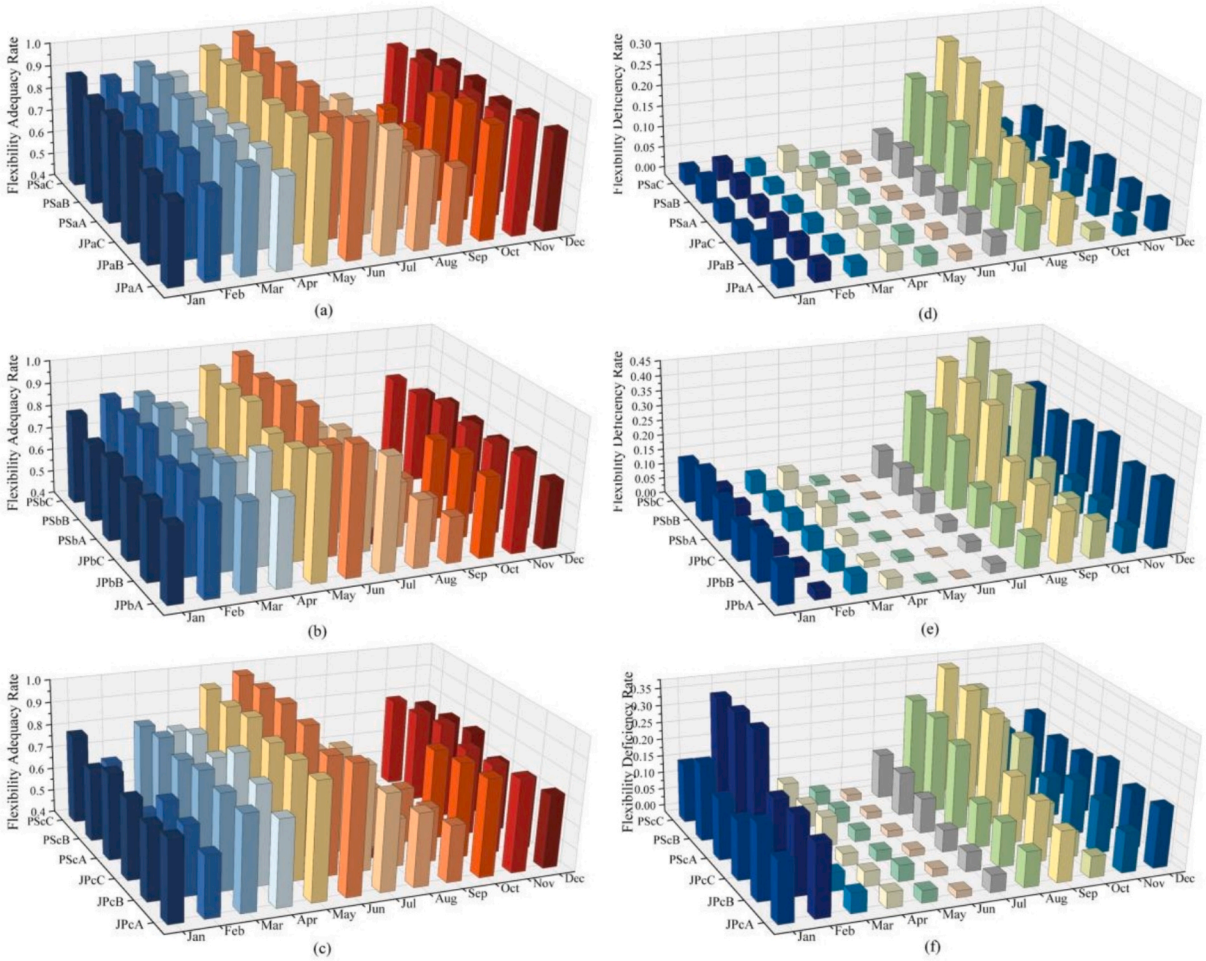


Fig. 11. Three-dimensional bar chart of the flexibility indicator.

in Low-Wind and Low-Irradiance scenario JPaA, while escalating to 0.126 in High-Wind and High-Irradiance scenario PScC. Notably, the disparity in deficiency rates between the two technologies peaks in specific months. As shown in Fig. 11(e), CPS (PSbC scenario) exhibits a deficiency rate of 44.6 % in October, while SPS (JPbC scenario) achieves 19.6 %–55.9 % reduction. In Fig. 11(f), CPS (PScC scenario) demonstrates significantly higher deficiency rates in February (34.0 %) and September (35.8 %) compared to SPS (JPcC scenario) (20.4 % and 18.0 %, respectively). The impact of wind-solar intensity on deficiency rates is pronounced: with increasing wind speeds, the deficiency rate rises from 0.026 in SPS (JPaA scenario) (January) to 0.161 in SPS (JPcA scenario) (January)–519.2 % surge. Similarly, under heightened irradiance, the rate increases from 0.070 in SPS (JPaA scenario) (August) to 0.093 in SPS (JPaC scenario) (August), representing 32.9 % growth. These findings collectively demonstrate that wind speed variations exert a greater impact on system flexibility than irradiance changes, and SPS exhibits distinct advantages in mitigating FDR under high wind-solar output conditions.

4.2. Economic viability comparison

To comprehensively evaluate the economic differences between SPS and CPS, this section calculates CAP and operational costs across all scenarios. Results summarized in Table. 3.

Table. 3 reveals distinct cost structure differences between SPS and CPS. SPS requires larger upper reservoirs and extended pressure tunnels, resulting in a significantly higher initial investment of 10,569.3 million (CNY) compared to CPS at 7,966.234 million. The construction costs for SPS are approximately 1.33 times those of CPS. However, SPS demonstrates a distinct economic advantage in terms of operational costs. Under JPcA, SPS achieves the lowest operational cost at 16.945 billion, whereas CPS (PScA scenario) under identical wind-solar conditions incurs substantially higher operational costs of 86.563 billion–approximately 5.1 times greater than SPS. A ranking analysis of operational costs across all scenarios reveals the findings: PSaA > PSaB > PSaC > PSbA > PSbC > PScA > PSbB > PScC > PScB > JPaA > JPaB > JPaC > JPbA > JPbC > JPbB > JPcB > JPcC > JPcA. Across all scenarios, CPS exhibits higher operational costs than SPS. This cost structure highlights that while SPS requires higher upfront capital investment, its optimized dispatch strategies and more efficient energy conversion significantly reduce long-term operational costs, demonstrating potential economic advantages in power systems with high wind-solar penetration.

While the total operational cost of SPS is demonstrably lower than that of CPS, its initial investment cost is approximately 1.33 times higher. To conduct a more comprehensive and systematic comparison of the lifecycle economics of both technologies, this study calculated their LCOE and performed a sensitivity analysis on capital cost uncertainty. The capital cost was varied within a range of 80 % to 120 % of its baseline value, with the results visually presented in the heatmap in Fig. 12.

Analysis of Fig. 12 reveals that the LCOE of SPS is highly sensitive to changes in its construction cost, exhibiting a clear linear growth relationship. The average LCOE for SPS is 0.58 CNY/kWh, higher than the 0.52 CNY/kWh average for CPS. This indicates that the advantage of SPS in operational costs is insufficient to offset its high initial investment, resulting in a higher levelized cost of electricity under most conditions

Table 3
Operating costs and construction costs for 18 scenarios (unit: 10000/CNY).

scenario	capital cost	operating cost	total system cost
JPaA	10,569,300	76,730	10,646,030
JPaB	10,569,300	70,422	10,639,722
JPaC	10,569,300	66,541	10,635,841
PSaA	7,966,232	309,196	8,275,428
PSaB	7,966,233	289,829	8,256,062
PSaC	7,966,234	282,699	8,248,933
JPbA	10,569,300	24,404	10,593,704
JPbB	10,569,300	21,754	10,591,054
JPbC	10,569,300	22,666	10,591,966
PSbA	7,966,232	96,052	8,062,285
PSbB	7,966,233	85,273	8,051,506
PSbC	7,966,234	91,688	8,057,922
JPcA	10,569,300	16,945	10,586,245
JPcB	10,569,300	17,036	10,586,336
JPcC	10,569,300	16,977	10,586,277
PScA	7,966,232	86,563	8,052,796
PScB	7,966,233	83,860	8,050,093
PScC	7,966,234	84,128	8,050,362

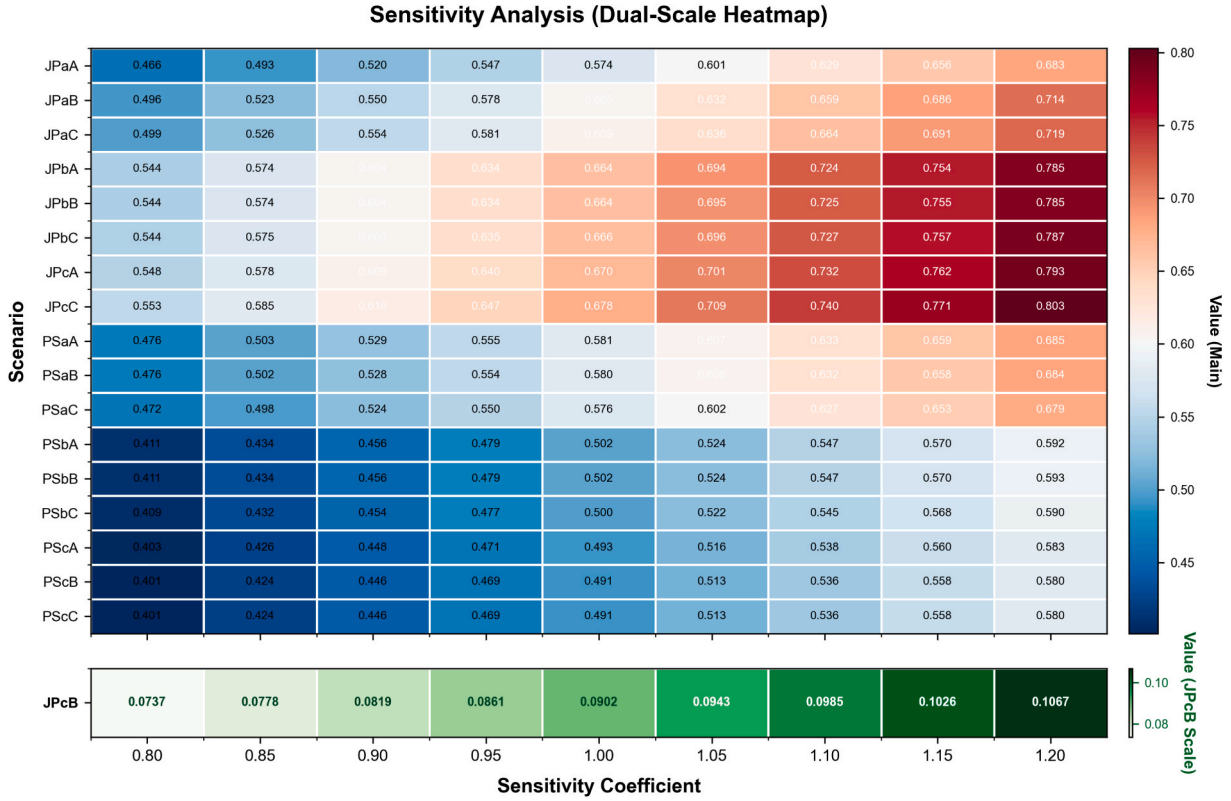


Fig. 12. Sensitivity Analysis of LCOE to Capital Cost (Calculation parameters: discount rate 8 %, project lifetime 50 years).

throughout its lifecycle. Notably, in the JPcB scenario (high-wind, normal-irradiance), the LCOE of SPS is exceptionally low and exhibits low sensitivity to changes in investment cost. This phenomenon is closely related to the highest net present value of electricity generation calculated for this scenario. Conversely, under the JPcC (high wind and high irradiance) scenarios, the LCOE of SPS peaks at 0.8 CNY/kWh,

approximately 37 % higher than that of CPS under equivalent conditions. This suggests that in scenarios with extreme renewable energy abundance, the operational cost savings achieved by SPS through frequent charging and discharging cycles are insufficient to compensate for its high capital expenditure. Although the LCOE of SPS is generally less competitive, the gap between the two technologies narrows significantly under low wind and low irradiance scenarios, where the LCOE of CPS even slightly exceeds that of SPS (by approximately 1 %). This indicates that under conditions of low renewable energy output and low system flexibility demand, the advantage of SPS's lower operational costs becomes relatively more pronounced [61].

In summary, despite the significant potential of SPS for reducing system-wide operational costs, its relatively high capital expenditure results in a lack of competitiveness for its own levelized cost of electricity in most scenarios.

4.3. Operational stability analysis of unit

Two key indicators were analyzed to evaluate the stability and reliability of SPS and CPS under stochastic wind-solar fluctuations: RGM and PAF. A higher RGM indicates stronger flexibility regulation capacity, while a higher PAF reflects lower operational stability. In Fig. 13, the left vertical axis represents RGM, while the right denotes the PAF.

In Fig. 12, SPS exhibits significantly higher RGM (with a mean of approximately 765 MW) compared to CPS (397 MW), while maintaining a lower regulation frequency. This demonstrates SPS's superior large-capacity, long-cycle regulation capabilities and enhanced operational stability. Under intensified wind-solar fluctuations, SPS exhibits a more pronounced increase in RGM alongside a relatively slower rise in regulation frequency. In contrast, CPS, despite lower RGM, shows a significant surge in regulation frequency (approximately 2,750–4,770 times), indicating its suitability for delivering frequent but smaller-magnitude regulation services. This divergence primarily stems from SPS leveraging larger reservoir capacities to facilitate long-duration, large-scale regulation, whereas CPS relies on its rapid start-stop cycling capabilities to deliver short-term, frequent adjustments.

Under high renewable generation volatility, load adjustment alone cannot satisfy system flexibility demands. Pumped storage units must consequently engage in frequent start-stop cycling and operational mode transitions to provide sufficient power regulation capacity. However, increased cycling frequency directly correlates with decreased operational stability. To quantitatively compare the flexibility of SPS and CPS, we analyzed their turbine-mode and pumping-mode start-up counts across all scenarios, as presented in the comparative bar charts of Fig. 14.

In Fig. 13, SPS exhibits higher turbine-mode and pumping-mode start-up counts compared to CPS in six scenarios: JPaA, JPaB, JPaC, PSaA, PSaB, and PSaC. This indicates low operational stability for SPS under these specific wind-solar conditions. In JPbA, JPbB, JPbC, PSbA, PSbB, and PSbC, CPS exhibits higher start-up counts than SPS, indicating reduced operational stability for CPS under these wind-solar conditions. In JPcA, JPcB, JPcC, PScA, PScB, and PScC, the start-up counts of SPS in JPcA are lower than that of CPS in PScA. Conversely, SPS in JPcB and JPcC exhibit higher start-up counts compared to CPS in PScB and PScC. As wind capacity and solar irradiance levels increase, both SPS and CPS exhibit a gradual upward trend in operational activity.

4.4. Comprehensive evaluation analysis

This section establishes a multi-dimensional evaluation framework using the integrated entropy-weighted method and coefficient of variation method. This framework holistically assesses the comprehensive performance of SPS and CPS under stochastic wind-solar power fluctuations. By applying a composite weighting approach to three key dimensions-technical performance, economic viability, and operational stability to objectively evaluate the adaptability and systemic value of both technologies in power systems with high renewable energy penetration.

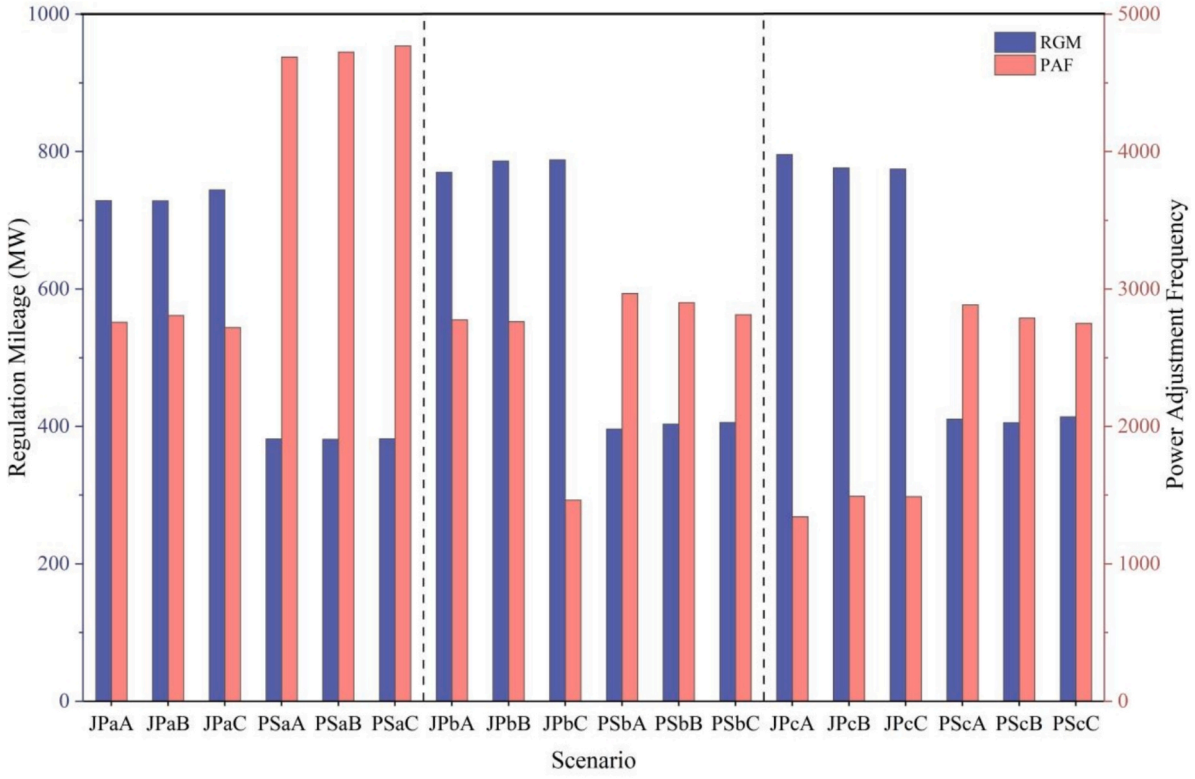


Fig. 13. Adjustment mileage and unit average power adjustment times for 18 scenarios.

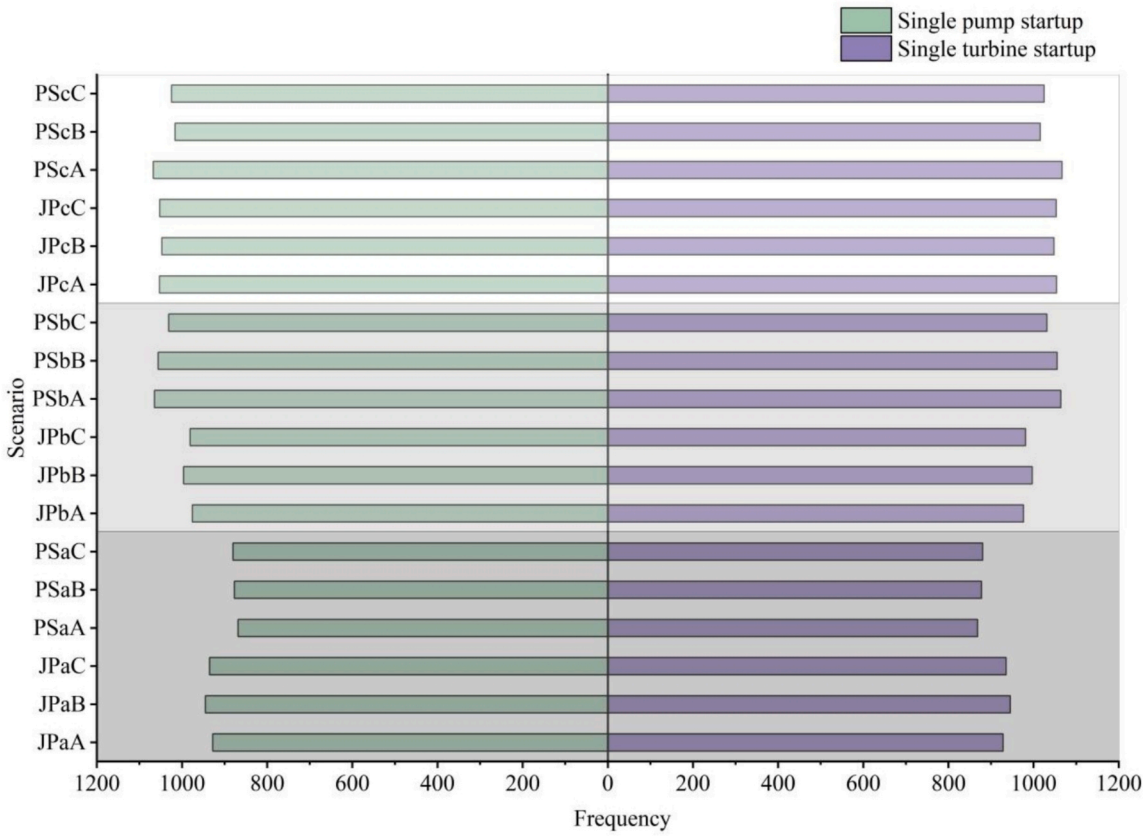


Fig. 14. Start-up times of single turbine and pump for 18 scenarios.

As shown in Table 4, four key indicators: FDR, CTR, NPD, and average power RGM exhibit gradual increases across all scenarios with rising wind speeds and irradiance levels. Notably, scenarios with SPS demonstrate significantly lower values in these indicators compared to those with CPS, along with relatively smaller growth magnitudes. In terms of CEM, scenarios incorporating SPS exhibit annual CEM hundreds of times lower than those with CPS. For example, the SPS scenario JPaA has a maximum emission of 12.51 million metric tons, whereas the C PS scenario PSaA under identical conditions reaches 206.3 million metric tons. Furthermore, CEM across all scenarios shows a significant downward trend with increasing wind capacity and solar irradiance levels, with SPS (JPcA, JPcB, JPcC scenarios) achieving zero CEM. Regarding stability indicators (start-up counts and power regulation frequency), scenarios with SPS show marginally higher values than those with CPS, though the annual difference remains within 100 cycles. However, SPS demonstrates a clear advantage in RGM, achieving 1.9 times the capacity of CPS. This suggests that SPS has high regulation capacity at the expense of stability, resulting in superior operational stability for CPS.

Fig. 15 illustrates the weight distribution across indicators in this composite evaluation model. The inner ring represents entropy-weighted-derived weights, the middle ring displays the coefficient of variation calculated weights, and the outer ring denotes the final composite weights derived from the combined approach.

It reveals distinct weighting priorities: entropy-weighted emphasizes

Table 4
Calculation results of all indicators for 18 scenarios.

Indicators	Technical						Stability		Economic	
Scenario	FAR	FDR	CTR	NPD (MW)	CEM (ton)	USS	PAF	RGM (MW)	OPC (10K)	CAP (10K)
JPaA	86.03%	3.04%	3.24%	10108	1.251E+08	1856	2758	728.79	76730	10569300
JPaB	85.54%	3.83%	3.47%	10422	9.301E+07	1890	2808	728.26	70422	10569300
JPaC	85.55%	3.98%	3.74%	10506	9.004E+07	1871	2720	744.41	66541	10569300
PSaA	79.60%	5.38%	6.28%	10797	2.063E+09	1737	2344	381.80	309196	7966232
PSaB	77.86%	6.21%	7.02%	11227	1.979E+09	1755	2362	381.30	289829	7966233
PSaC	77.16%	6.77%	7.39%	11232	1.999E+09	1760	2385	382.19	282699	7966234
JPbA	79.68%	8.51%	4.39%	10591	9.712E+06	1952	2776	769.88	24404	10569300
JPbB	80.32%	9.42%	5.70%	10902	5.827E+06	1993	2763	786.42	21754	10569300
JPbC	78.38%	10.34%	6.05%	10968	6.977E+06	1962	2928	788.03	22666	10569300
PSbA	69.44%	13.95%	11.07%	11045	5.879E+08	2129	2968	396.31	96052	7966232
PSbB	68.72%	15.08%	11.83%	11454	5.222E+08	2111	2902	403.36	85273	7966233
PSbC	68.17%	16.03%	12.29%	11494	5.802E+08	2062	2814	405.79	91688	7966234
JPcA	79.54%	8.26%	4.63%	10267	0.000E+00	2107	2686	795.84	16945	10569300
JPcB	78.66%	9.74%	5.30%	10577	0.000E+00	2096	2984	776.46	17036	10569300
JPcC	80.07%	9.88%	5.47%	10616	0.000E+00	2106	2980	774.67	16977	10569300
PScA	68.56%	14.77%	11.14%	10724	4.905E+08	2134	2885	410.53	86563	7966232
PScB	66.34%	16.02%	11.87%	11006	4.856E+08	2032	2789	405.64	83860	7966233
PScC	66.62%	16.72%	12.31%	11081	4.890E+08	2049	2751	413.98	84128	7966234

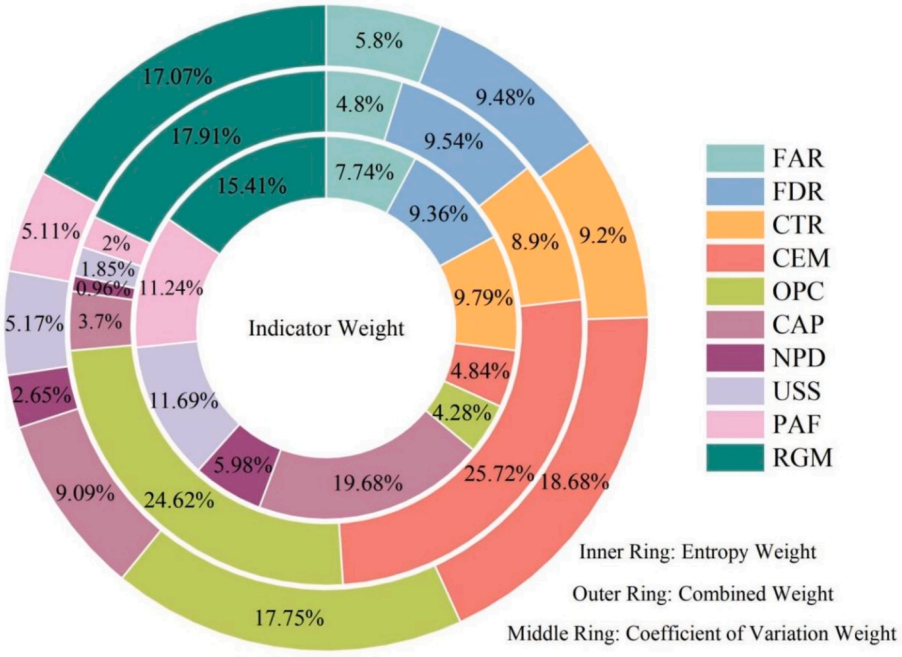


Fig. 15. Weight coefficient of each evaluation index.

construction costs and average RGM, with a combined weight of 35.09 %. Conversely, the coefficient of variation prioritizes CEM (25.72 %) and operational costs (24.62 %). In the final composite weighting system, the carbon emission indicator holds the highest weight at 18.68 %, followed by the operational cost indicator (17.75 %) and average RGM (17.07 %). These three core indicators represent the technical, economic, and stability dimensions, respectively. Comprehensive calculations indicate that, within the combined weighting system based on the entropy-weighted and coefficient of variation methods, technical indicators account for 45.81 %, economic indicators for 26.84 %, and stability indicators for 27.35 %. This suggests that the constructed comprehensive benefit evaluation model places greater emphasis on the impact of technical indicators on the overall system performance.

Based on the weights of each indicator, the comprehensive scores for

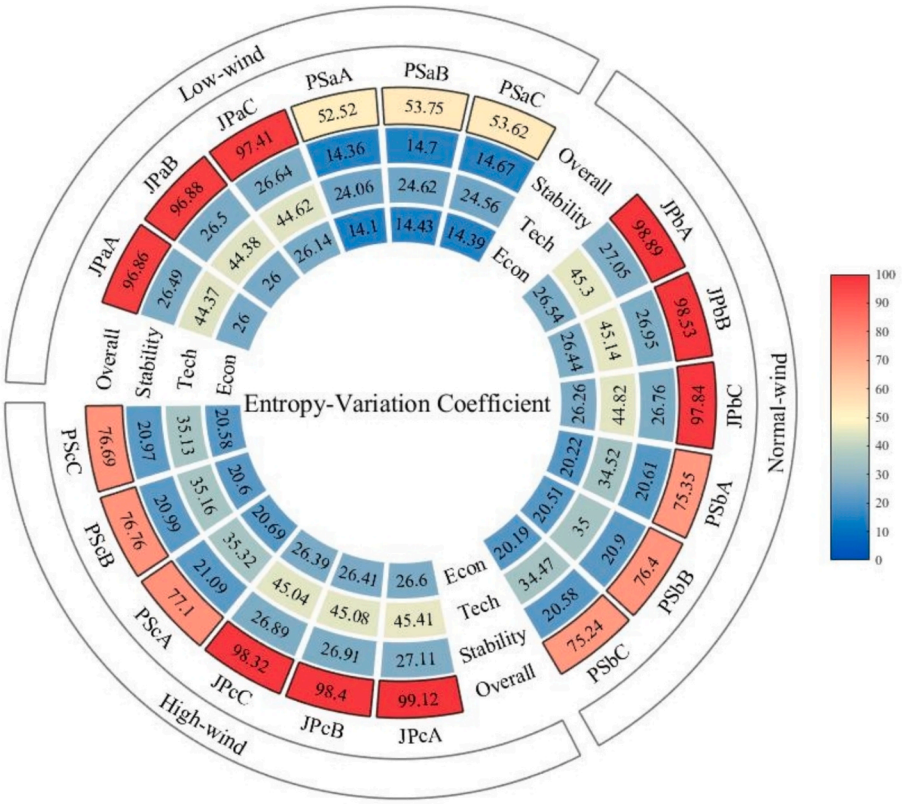


Fig. 16. Comprehensive benefit evaluation results.

18 scenarios were calculated. In Fig. 16, the layers from the center outward represent, in order, the technical, economic, and stability scores, followed by the overall score for each scenario. The color intensity in the figure indicates the relative score, with darker shades representing higher scores.

The Fig. 16 provides a multi-dimensional visualization of the comprehensive scores. The chart's layered architecture allows for a simultaneous comparison across different dimensions: moving from the center outward, the concentric layers correspond to the aggregated scores for the technical, economic, and stability dimensions, respectively. The outermost boundary of each segment represents the final, overall comprehensive score for that specific scenario. The color intensity is proportional to the score's magnitude, with darker shades indicating superior performance, which clearly highlights the comparative advantages of different strategies.

It shows that scenarios incorporating SPS have an average comprehensive benefit score of approximately 98.03, while those with CPS average around 68.6, which is significantly lower than SPS. Among the scenarios with SPS, the top five in terms of comprehensive scores are JPcA (99.12), JPbA (98.89), JPbB (98.53), JPcB (98.4), and JPcC (98.32). JPcA achieved the highest scores across all three categories: economic, technical, and stability indicators. Furthermore, as wind and irradiance increase, the scores for all three indicators show an upward trend. It indicates that SPS demonstrates excellent overall performance in scenarios with high wind and irradiance.

5. Conclusion

This study comprehensively evaluates the regulatory performance of SPS and CPS systems under inherent wind and solar power fluctuations. By simulating diverse combined scenarios of typical wind and solar years, it identifies their distinct regulatory capabilities across a spectrum of renewable energy conditions. Key conclusions are summarized:

(i) Regarding technical performance, CTR, NPD, and FDR exhibit a gradual upward trend with increasing wind and irradiance across all scenarios, while CEM and FAR show declining trends. SPS (JPaA) achieves the best performance in CTR, NPD, FAR, and FDR. In terms of CEM, the JPcA, JPcB, and JPcC scenarios achieve zero CEM. These results suggest that SPS may deliver stronger carbon emission suppression effects, especially under higher wind energy conditions. Meanwhile, SPS significantly improves the regulation of net load fluctuations, reducing the peak-to-valley difference by 10.6 % compared to CPS in high wind and solar scenarios. This underscores the effectiveness of SPS's unique inter-seasonal energy storage capability in managing large-scale fluctuations of renewable energy. Regarding stability indicators, SPS demonstrates a clear advantage in RGM, achieving 1.9 times that of CPS. This indicates that while SPS offers high regulation capacity, it may do so at the expense of some stability, whereas CPS maintains relatively superior operational stability.

(ii) Economically, while SPS requires an initial investment approximately 1.33 times higher than CPS, its operating costs are dramatically lower. However, a comprehensive LCOE analysis reveals that SPS's elevated capital expenditure leads to an average LCOE of 0.58 CNY/kWh, which exceeds CPS's 0.52 CNY/kWh. This indicates that SPS's operational cost advantages are currently insufficient to fully offset its substantial upfront investment requirements, resulting in a higher lifecycle cost under most conditions. The comprehensive evaluation shows that when the weights of technical, economic, and stability indicators are 45.81 %, 26.84 %, and 27.35 %, respectively, the average comprehensive benefit score for scenarios with SPS is about 98.03, while that for scenarios with CPS is about 68.6, which is significantly lower than SPS. And the top five in terms of comprehensive scores are JPcA (99.12), JPbA (98.89), JPbB (98.53), JPcB (98.4), and JPcC (98.32). It indicates that SPS demonstrates excellent overall performance in scenarios with high wind and irradiance.

(iii) The findings yield critical policy implications for the energy transition. To support its deployment, it is essential to design market mechanisms, such as capacity markets or long-term contracts, that can appropriately compensate the long-duration flexibility services provided by SPS and mitigate its high upfront investment risks. Ultimately, energy planners should consider proactively integrating SPS into strategic long-term planning, such as Integrated Resource Plans, using quantitative evidence from comprehensive assessments to ensure that future grid expansion is both cost-optimal and resilient.

6. Dicussion

(i) Geographical Limitations and Global Applicability of SPS Findings. This study offers a comprehensive evaluation of SPS and CPS performance under variable renewable energy scenarios. However, a primary limitation is its geographical constraint to the Qinghai region, which may limit the generalizability of findings. The superior performance of SPS in high wind and solar irradiation scenarios may vary in regions with different climate patterns, load characteristics, and resource endowments. Nevertheless, this study's findings and methodological framework remain highly relevant for numerous other regions globally grappling with similar energy transition challenges. The superiority of SPS is particularly pronounced in regions characterized by three concurrent factors: (1) high penetration levels of variable renewable energy (VRE); (2) a significant seasonal mismatch between VRE generation profiles and electricity demand; and (3) suitable topographical and hydrological conditions for large-scale reservoir construction. Further research should extend the geographical scope to validate these findings across diverse global contexts.

(ii) Comparative Advantages and Disadvantages of CPS and SPS Technologies. Conventional Pumped Storage (CPS) remains a mature and reliable technology, with its primary advantages being a lower relative capital cost and high efficiency in providing short-term, intra-day flexibility and ancillary services. It is an effective solution for daily peak shaving and frequency regulation. However, its fundamental limitation, as highlighted in this study, is its inability to address the large-scale, seasonal energy imbalances inherent in VRE-dominated systems, which can lead to higher system-wide operational costs and increased mechanical stress on units. Seasonal Pumped Storage (SPS), conversely, offers an unparalleled capability for inter-seasonal energy arbitrage, which is its core advantage. This allows it to drastically reduce long-term operational costs, minimize renewable curtailment on a seasonal scale, and enable deep decarbonization of the power system. The primary disadvantages of SPS, however, are its extremely high upfront capital investment, long construction lead times, and its dependence on very specific topographical and hydrological conditions, making suitable sites much rarer than for CPS.

(iii) Hybrid Operational Strategies and Value Stacking for SPS. This study did not explore hybrid operational strategies for SPS. While an SPS plant is technically capable of operating on a daily cycle to provide ancillary services similar to a CPS, this would likely be a suboptimal use of the asset. The primary value proposition of SPS is its unique ability to perform inter-seasonal energy arbitrage, which typically offers far greater economic returns than intra-day markets. Future research should therefore focus on developing sophisticated co-optimization models to quantify the potential revenue streams from stacking both long-term energy arbitrage

and short-term ancillary services, thereby determining the true optimal operational strategy for these versatile assets.

CRediT authorship contribution statement

Meng Zhang: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation. **Peiquan Li:** Writing – review & editing, Resources, Data curation. **Yaling Yao:** Writing – review & editing, Resources, Data curation. **Ai Relanjiang Kanji:** Writing – review & editing, Visualization, Resources. **Ziwen Zhao:** Writing – review & editing, Visualization. **Diyi Chen:** Writing – review & editing, Software, Resources. **Natália De Assis Brasil Weber:** Writing – review & editing, Resources. **Julian David Hunt:** Writing – review & editing, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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